

Unequal Opportunities: Worker Adjustment After a Sectoral Collapse

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Abstract

Displaced workers in the same local labor market, facing the same sectoral shock and observationally similar, return to work at sharply different rates. I show that this heterogeneity is predicted by skill–demand alignment: the match between a worker’s pre-shock skills and the composition of surviving local labor demand, summarized in a worker-level reallocation index constructed from pre-shock administrative data. Studying Spain’s 2007–2012 construction collapse, I find that displacement costs operate predominantly through extended non-employment, that adjustment runs through sectoral rather than geographic mobility, and that moving from the bottom to the top quartile of the index reduces cumulative employment losses by 34 percent. The index predicts how quickly workers return to work but not the wages they earn once reemployed—consistent with the index governing offer arrival rather than match quality. Employment recovery is a property of the worker–market pairing; earnings losses additionally depend on the sectoral structure into which workers reallocate. Neither worker-based nor place-based targeting alone reaches the relevant margin.

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1 Introduction

Workers displaced by a large sectoral contraction follow substantially different reemployment trajectories, even within the same local labor market and among workers with similar observable characteristics. The displacement literature largely abstracts from this variation. One strand models reallocation prospects as a function of worker characteristics such as human capital, tenure, and industry-specific skills, treating location as secondary (Jacobson et al., 1993; Gulyas and Pytka, 2025; Autor et al., 2014). Another emphasizes local labor-market conditions, treating workers in a region as facing a common set of opportunities (Yagan, 2019; Notowidigdo, 2020). Neither perspective explains the pronounced heterogeneity in adjustment outcomes among workers exposed to the same local shock, or identifies which workers within a distressed region bear the largest costs of structural change.

This paper argues that the relevant unit of analysis is neither the worker nor the local labor market in isolation, but the worker–market pairing: the interaction between a worker’s pre-shock skills and the demand structure of the surviving local economy, which I refer to as skill–demand alignment. The central question is who can successfully reallocate after a sectoral collapse. I show that the answer is largely predictable from administrative data observed before the shock occurs. Two findings organize the analysis. First, skill–demand alignment strongly predicts reemployment and sectoral mobility, but not wages conditional on reemployment—a pattern consistent with an offer-arrival interpretation, under which alignment governs access to acceptable opportunities rather than the terms of resulting matches, and difficult to reconcile with match-quality or unobserved-ability explanations. Second, the employment and earnings consequences of displacement reflect different sources of variation: the employment gradient is primarily a worker-level phenomenon governed by alignment, while the earnings gradient additionally reflects the sectoral composition of the local economy into which workers reallocate.

Skill–demand alignment lacks a measure with testable economic content. This paper’s primary contribution is to provide one. The measurement device is a worker-level index, $RI^w(X_i)$, constructed from pre-shock administrative data, that recovers worker-specific reallocation opportunities within a local labor market—an object neither literature separately identifies. A simple directed-search model in the tradition of Burdett and Mortensen (1998) motivates a single-index representation: a demand-weighted sum, across non-origin sectors, of whether a worker’s skills transfer, whether the sector is locally available, and whether switching pays (Section 4). What distinguishes the index from a generic opportunity measure is that it is an *arrival rate*: it predicts how fast a worker exits non-employment and how readily she switches sectors, but not the wage she commands once matched—a partition that

separates offer arrival from match quality and is the paper’s central discriminating device.

Framing the index as an offer-arrival rate disciplines the empirical work. The model predicts that non-employment duration and cumulative earnings losses both decline with the index—predictions many displacement models share. The discriminating test is the conditional-wage null: since the index governs how quickly acceptable offers arrive rather than the terms of resulting matches, it should not affect wages conditional on reemployment. An alternative in which high-index workers access structurally better-paying sectors would predict a positive conditional-wage effect instead. In the data, conditional wages exhibit no meaningful response to the index while the employment margin responds sharply—the pattern predicted by the offer-arrival interpretation, and the opposite of what a match-quality account predicts. This framing sharpens existing worker-level opportunity measures: the closest antecedent is the outside-options index of [Caldwell and Danieli \(2024\)](#), but that index is a portable worker attribute, whereas the reallocation index is a worker-in-province object, so two workers with identical skills face different prospects in provinces with different surviving demand ([Yi et al., 2024](#); [Macaluso, 2025](#)).

Spain’s construction collapse between 2007 and 2012 provides an unusually well-suited setting. Construction’s employment share fell from 13 percent to below 6 percent, remaining 59 percent below its 2007 peak as late as 2019. Because the decline reflected structural overbuilding rather than cyclical weakness, reallocation rather than recall was the relevant margin. The collapse was triggered by an international sudden stop ([Mendoza, 2010](#); [Bentolila et al., 2012](#)), so cross-province differences in exposure were largely determined by pre-recession specialization—predetermined with respect to post-shock outcomes. Construction employed workers across a broad skill range, so a single provincial shock generates markedly different reallocation opportunities across workers, while limited geographic mobility kept adjustment within local labor markets, producing the within-market variation the analysis exploits. Spain is unusual in combining these features with administrative data spanning the pre-boom period, allowing the index to be constructed from a predetermined economic structure.

Identification rests on a key property of the index: most variation occurs within rather than across provinces—approximately 2.3 times larger within than between. That within-province variation dominates is itself informative: if adjustment capacity were primarily a feature of local labor markets, most variation would appear across provinces. This variation is predetermined, reflecting career histories accumulated before the recession; pre-recession earnings trends are parallel across the index distribution conditional on province and industry, and main results are robust to instrumenting provincial exposure with pre-recession construction growth. The identifying assumption is that, conditional on shock severity and

pre-displacement characteristics, within-province index variation does not reflect unobserved worker quality that independently predicts post-displacement outcomes. Most directly, the shock–index interaction for employment is essentially unchanged when province fixed effects absorb all province-level confounds, confirming that the moderating role of alignment is identified from within-province cross-worker variation alone (Section 8).

The construction collapse generated large and persistent earnings losses, operating predominantly through extended non-employment: roughly three-quarters of the cumulative loss reflects reduced employment rather than lower conditional wages. Adjustment ran through sectoral reallocation rather than geographic relocation: the annual probability of leaving construction rises by approximately 5 percentage points at the interquartile shock difference, while interprovincial migration is statistically indistinguishable from zero; even after netting out correlated declines in potential destinations (Bertoli and Moraga, 2013; Borusyak et al., 2022), the sectoral response exceeds the geographic response by at least a factor of six.

Skill–demand alignment strongly predicts which workers bear the largest adjustment costs. Moving from the bottom to the top quartile of the index reduces cumulative employment losses by 34 percent and earnings losses by 35 percent, conditional on shock severity and pre-displacement characteristics. The two gradients have different sources: the employment gradient is governed by worker-level alignment, while the earnings gradient additionally reflects the local sectoral structure into which workers reallocate (Section 6). Low-index workers disproportionately reallocate into hospitality and agriculture—sectors with high turnover and limited earnings growth—while high-index workers transition into a broader range of destinations, including manufacturing, professional services, and finance, where employment spells are more stable and earnings prospects stronger.

The evidence supports an offer-arrival interpretation. Two discriminating tests support this reading. First, a wage-excluded version of the index—constructed solely from the overlap between pre-displacement skills and surviving local labor demand—predicts adjustment outcomes nearly as strongly as the baseline measure; a wage-only variant that weights local demand by sectoral wage premia performs poorly and enters with the wrong sign. Second, as noted above, the shock–index interaction is large and precisely estimated for employment while the corresponding interaction for conditional wages is consistently indistinguishable from zero (Section 6). Workers with stronger alignment return to work faster but do not earn higher wages once reemployed—inconsistent with match-quality or wage-setting accounts.

I show that heterogeneity in adjustment costs is systematically predictable from pre-shock data and reflects the alignment between workers’ skills and surviving local labor demand. I complement Beaudry et al. (2012), who show that local industrial composition shapes wages through outside options in steady state, by documenting that the same forces govern

employment and earnings trajectories following job loss.

The reallocation index makes the question of which workers bear the largest adjustment costs answerable using only pre-shock data. The Spanish collapse is one parameterization of a more general framework—defined by limited short-run geographic mobility and cross-regional variation in sectoral demand—rather than a setting to which the framework is specific.¹ Both features characterize manufacturing regions exposed to import competition (Autor et al., 2013; Dix-Carneiro and Kovak, 2017), energy-extraction communities, and labor markets reshaped by automation (Acemoglu and Restrepo, 2020). Because the index is constructed from matched worker–firm data observed before the shock—increasingly available across OECD countries—the measurement framework travels beyond the Spanish setting, even if quantitative magnitudes need not. The policy implication follows from the two-gradient decomposition: because the employment margin is a worker-level object, worker-targeted instruments can be aimed at the workers who will bear the largest losses using pre-shock data alone, while the earnings margin depends on local structure and remains the province of place-based policy.

The remainder of the paper proceeds as follows. Section 2 discusses the collapse of Spain’s construction sector. Section 3 describes the data, and Section 4 introduces the reallocation index. Section 5 documents the labor market consequences of the shock, while Section 6 shows how worker–demand alignment mitigates displacement costs and the underlying adjustment mechanisms, and Section 7 presents a counterfactual policy exercise. Section 8 reports robustness checks. Section 9 concludes.

2 Setting: Spain’s Construction Collapse

Between 2007 and 2012, Spain’s construction sector shed nearly 1.2 million workers, contracting from 13 percent of total employment to below 6 percent (Figure 1). The collapse appears structural rather than cyclical: Spain had built housing at rates far exceeding long-run demand during the boom, driven by interest rate convergence following EU accession and amplified by domestic credit expansion. Once credit conditions normalized, employment remained 59 percent below its 2007 peak as late as 2019. In contrast, the United States, where construction largely recovered by 2015 as an underlying housing shortage reasserted itself (Appendix Figure A4).

The Spanish setting has three properties that make it well-suited for this question. First, the shock originated in a sudden stop of international capital inflows (Mendoza, 2010; Ben-

¹These correspond to low geographic mobility and near-complete collapse of construction hiring; Appendix B.2 shows that relaxing either yields an equivalent measure.

tolila et al., 2012), so cross-province differences in severity trace to pre-recession sectoral specialization, plausibly predetermined with respect to workers’ post-2007 trajectories; measuring exposure in levels rather than shares yields nearly identical results (Table 1, Column 4). Second, interregional mobility in Spain is low, so adjustment occurs predominantly within local labor markets and the shock is not confounded by selective sorting across provinces. Third, construction spans the full skill distribution within a single sector and province, so the same provincial shock generates sharply heterogeneous reallocation prospects across workers, enabling within-province comparisons that hold local labor demand fixed. Section 5.3 and 6.2 provides complementary evidence that conditional on controls, the provincial variation in exposure is uncorrelated with pre-recession worker outcomes.

Figure 1: Employment share of workers in the Spanish construction sector, 2000-2021



Notes: Employment share of Spain’s construction sector, January 2000–December 2021. Monthly observations are based on workers aged 20–60. The shaded area denotes the Great Recession period (2007–2012).

The contraction exhibited pronounced regional heterogeneity (Appendix Figure A11): provincial construction employment losses ranged from around 20 percent in several northern provinces to more than 60 percent in the south, with an interquartile range of 10.3 percentage points in the provincial employment share decline.

Figure A1 tracks workers employed in construction in 2007 and illustrates the breadth of the displacement.² By 2015, roughly 30 percent of former construction workers were

²The patterns in Figure A1 are not driven by compositional differences across skill groups or by pre-existing sectoral mobility trends. Appendix Figure A12 replicates the analysis for high-skilled construction workers and finds qualitatively identical dynamics, though with smaller employment losses, confirming that the broad reallocation pattern is not specific to low-skill occupations. Appendix Figure A13 applies a placebo

employed in other sectors, while the share who relocated across provinces remained below 8 percent.

3 Data

The primary data source is the *Muestra Continua de Vidas Laborales* (MCVL), a longitudinal administrative dataset comprising a 4 percent random sample of Spain’s Social Security affiliates. The MCVL links Social Security records with income tax filings and the population register, providing longitudinal labor market histories. Spell-level records cover employment and unemployment episodes, including earnings, contract type, firm size, two-digit industry, occupation, part-time status, workplace location, and reason for job separation.³ Individual characteristics include gender, educational attainment, date and province of birth, citizenship, and household composition.

Earnings are available from two sources. Social Security records provide monthly earnings back to 1980 but are subject to annually adjusted contribution floors and ceilings; tax records, linked from 2006 onward, are uncensored. Censoring is minimal in the construction sample—only 0.6 percent of worker-month observations are top- or bottom-coded—so I use tax records as the primary measure from 2006 onward.⁴ The uncensored tax records permit analysis across the full earnings distribution. All earnings are deflated to 2009 prices using the national CPI.

I construct a monthly panel spanning 2000–2019. The complete sample includes all workers aged 18–60 affiliated with Social Security across all 50 Spanish provinces, excluding Ceuta and Melilla, and is used for descriptive analysis.⁵ The estimation sample applies four restrictions. First, at least six months of construction employment during 2005–2006, targeting workers with established sectoral attachment (average construction tenure in 2007 was 11 months) and excluding marginal spells from the boom’s final expansion.⁶ Second, native workers only: foreign-born workers constituted 27 percent of construction employment in 2007 and exhibit higher rates of return migration during recessions, generating differential

shock in 2003 and finds no differential pre-period trajectories, ruling out the interpretation that Figure A1 reflects pre-recession differences in sectoral composition rather than adjustment to the collapse itself.

³Contract information is recorded at the spell level, including exact start and end dates.

⁴Bonhomme and Hospido (2017) document that earnings differences between tax and Social Security records are concentrated above the 90th percentile. Workers in the Basque Country and Navarre are covered by Social Security records only due to separate fiscal arrangements. I compute days worked within each calendar month from the underlying employment spells and convert annual tax earnings into daily equivalents to ensure consistency across sources.

⁵Ceuta and Melilla are excluded due to their distinct labor market structures and fewer than 200 construction workers in 2007.

⁶Results are robust to alternative thresholds of 3, 9, and 12 months (Appendix Table A10).

attrition in high-exposure provinces not observable in the administrative records.⁷ Third, age 20–50 in 2007, to isolate labor market adjustment from retirement transitions. Workers are followed from January 2007 through December 2019, with the main analysis ending in December 2012. The resulting estimation sample comprises 34,000 workers; summary statistics are reported in Appendix Table A1.

4 The Reallocation Index

Workers displaced by the same shock do not face the same prospects. A bricklayer and an electrician laid off in the same province have access to different sets of potential jobs, because their skills transfer differently across sectors and because local labor demand is distributed unevenly across activities. This worker-specific opportunity set is the object this paper measures. Conventional regional designs, which assign a single exposure measure to everyone in a province, do not capture it: they absorb differences in skill–demand alignment into aggregate local conditions. This section builds an index of that opportunity set, explains the intuition behind its form, and states the predictions that organize the empirical analysis; Appendix B.1 provides the formal framework.

4.1 A Simple Framework

Consider a worker searching across local sectors after losing a construction job. Whether she returns to work quickly depends on how often she receives an offer she is willing to accept. A standard directed search framework (Appendix B.1) implies that acceptable offers arrive more frequently when three conditions hold: a worker’s skills are compatible with a potential destination sector, that sector is well represented in the local labor market, and that sector offers wages that are attractive relative to pre-displacement earnings.⁸ These three forces motivate combining skill compatibility, local opportunity, and wage adequacy into a single multiplicative index.

Let X_i denote a vector of worker i ’s observable characteristics—occupation, foreign-born status, education, age, and gender—and r her province of residence. The wage-weighted reallocation index is

$$\text{RI}^w(X_i, r) = \sum_{j \neq \text{cs}} \hat{\phi}_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{\text{cs}}}, \quad (1)$$

where $\hat{\phi}_j(X_i)$ is the predicted probability that a worker with characteristics X_i is employed

⁷Results are robust to controlling for baseline foreign-worker shares (Section 8).

⁸Formally, labor-market tightness is assumed to increase with local sectoral specialization, and offer acceptance to increase with the relative wage. These restrictions are detailed in Appendix B.1.

in sector j ; $s_{jr}/\bar{s}_j$ is the relative concentration of sector j in province r ; and w_j/w_{cs} is the ratio of sector j 's median wage to the construction-sector median wage, with all measured in 2006. The measure, $RI^w(X_i, r)$ (hereafter RI_i), summarizes the earnings-weighted alignment between a worker's skill profile and the structure of local labor demand.

Each component captures one of the three forces above. The match probability $\hat{\phi}_j(X_i)$ stands for skill compatibility: the share of sector- j openings a worker like i could plausibly fill. The specialization ratio $s_{jr}/\bar{s}_j$ stands for local opportunity: it gives more weight to sectors that are overrepresented in the worker's province, while preventing large national sectors from dominating mechanically; it excludes construction's own local share, so the index measures prospects *out of* the declining sector. The wage ratio w_j/w_{cs} stands for wage adequacy: offers paying far below pre-displacement earnings are less likely to be accepted and so contribute less.⁹

All three inputs are measured before the recession, ensuring each component is predetermined with respect to post-2007 outcomes. Match probabilities are estimated from a multinomial logit of sectoral employment on hiring spells of non-construction workers between 2002 and 2006, so the probabilities reflect sector entry opportunities rather than incumbency patterns. Local sectoral shares s_{jr} are measured at the province level in 2006, excluding construction; sector median wages w_j are computed nationally from 2006 earnings.¹⁰

Two concerns about interpretation deserve attention. First, whether the skill-sector mapping continued to operate during the downturn. Because $\hat{\phi}_j(X_i)$ reflects relatively stable structural features of how worker characteristics match sector requirements, it should be largely unaffected by the contraction in employment levels. Consistent with this, re-estimating the logit on post-recession hiring spells (2010–2015) yields an index highly rank-correlated with the baseline ($\rho = 0.873$). Second, sorting: if high-ability workers sort into provinces with favorable sectoral composition, the index might absorb unobserved quality rather than local opportunity. Conditioning on observable characteristics addresses the component of sorting the index summarizes, and the abruptness of the capital-flow reversal makes anticipatory sorting implausible (Section 2). Most directly, the index has no predictive power for *pre-recession* employment trajectories (Figure A2), as one would expect if it captures structural opportunity rather than unobserved worker quality.

⁹The wage ratio proxies for the probability that an offer is accepted, not for match surplus. A uniform-weight variant that replaces w_j/w_{cs} with unit weights therefore ignores variation in the attractiveness of potential offers. Section 6.3 shows that the predictive content of the index comes primarily from skill transferability rather than differential access to high-wage sectors.

¹⁰National rather than province-specific wages capture sector-specific skill requirements common across regions and avoid local cost-of-living differences unrelated to job quality.

The model delivers asymmetric predictions across the two margins of adjustment. The offer arrival rate depends on within-province sectoral shares, so its cross-worker variation is identified independently of province-level wages; the expected reemployment wage, by contrast, depends on the absolute wage level of destination sectors in province r . Worker-level alignment should therefore predict reemployment speed within a province after absorbing all province-level variation, while the earnings gradient partially reflects between-province differences in the wage level of surviving sectors. Section 6.3 confirms both predictions.

4.2 Properties: Within- versus Between-Province Variation

The index varies along two dimensions. Across provinces, the same worker profile maps into different opportunity sets as local sectoral composition changes; within a province, different worker profiles access different subsets of the same local labor market. The within-province dimension dominates: the within-province standard deviation of the index is 2.3 times the between-province standard deviation in the estimation sample.

The baseline index isolates the margin that binds most tightly in the Spanish setting: when workers cannot relocate and the origin sector has ceased hiring, skill–demand alignment within the local labor market is the primary adjustment channel. Two extensions in Appendix B.2 relax these restrictions. When geographic mobility is feasible, access to neighboring labor markets partially substitutes for local alignment. When the contracting sector retains residual demand, the index converges toward a conventional outside-options measure. Both extensions clarify the scope conditions under which the baseline index captures the relevant opportunity set.

Workers with greater reallocation opportunities should return to work more quickly, experience smaller earnings losses, and sort into sectors compatible with their skills and well-represented locally. A further implication is that aggregate measures of sectoral diversity should contain little additional information once worker-specific opportunities are accounted for; Section 6.3 assesses this prediction directly.

5 The Cost of the Construction Collapse

5.1 Empirical Strategy and Identification

Identification exploits cross-province variation in exposure to the 2007–2012 construction contraction. I compare observably similar workers in provinces that experienced differential declines in construction activity, using the decline in provincial construction employment share as the shock measure. The employment share decline is interpretable as a labor demand

shock: regional employment fluctuations are predominantly driven by demand rather than supply in the medium run (Blanchard and Katz, 1992), and the trigger here, an abrupt tightening of international credit conditions beginning in 2007–2008, was common across provinces (Section 2). Province-level variation in severity therefore reflects pre-recession differences in sectoral composition rather than worker behavior or local labor supply.

Because adjustment occurred over several years, I focus on cumulative earnings over 2007–2012, normalized by average annual earnings in 2005–2006 (Autor et al., 2014; Yagan, 2019). This measure captures the combined effects of wage changes and non-employment over the full adjustment period. The baseline specification is:

$$y_i = \theta \text{Shock}_{r(i)} + X_i' \beta + \varepsilon_i, \quad (2)$$

where $\text{Shock}_{r(i)}$ denotes the proportional decline in construction’s employment share between 2007 and 2012 in worker i ’s pre-recession province of residence, r .¹¹ The vector X_i contains predetermined worker characteristics measured in 2007, including age interacted with gender and education, occupational skill group, tenure, labor-market experience and its square, indicators for fixed-term and part-time contracts, and baseline earnings quintiles. It also includes province-level controls measured in 2006: the unemployment rate, construction’s initial employment share, a Bartik demand shifter for non-construction sectors, and a Herfindahl-Hirschman index of non-construction sectoral concentration.

As a complementary exercise, I estimate an IV specification, treating IV results as corroborative rather than primary because instrumenting the shock substantially reduces statistical power for the heterogeneity analysis that is the paper’s main focus. I instrument the provincial shock with average annual growth in construction employment during 2000–2006: provinces with faster pre-recession expansion accumulated greater sectoral exposure, making them more vulnerable to the subsequent collapse. The instrument is informative. A one-percentage-point increase in pre-recession construction growth predicts a 1.287-percentage-point larger decline in the construction employment share between 2007 and 2012 (Appendix Figure A8), yielding a Sanderson–Windmeijer F-statistic of 12.44. Moreover, weak-instrument-robust Anderson–Rubin inference rejects the null of no effect, with a 95 percent confidence interval of $[-6.08, -1.27]$ ($p = 0.017$).¹²

The main threat is that pre-recession construction growth proxies for broader provincial characteristics correlated with post-recession outcomes. Two pieces of evidence address

¹¹Let s_{rt} denote construction’s share of total employment in province r and year t . The shock is defined as $-(s_{r,2012} - s_{r,2007})/s_{r,2007}$, such that larger values indicate a more severe construction collapse.

¹²Wild-bootstrap inference yields similar conclusions (see Section 8), addressing concerns associated with inference based on 50 province clusters (Cameron et al., 2008).

this. First, the first-stage coefficient is virtually unchanged when controlling for pre-recession growth in mortgage originations, mortgage values, and immigrant population (Table A4), indicating the instrument does not simply capture differential credit, housing-market, or labor-supply dynamics. Second, regressions of cumulative earnings, annual earnings, and cumulative days worked during 2003–2006 on the instrument yield small, statistically insignificant coefficients across all three outcomes (Table A3), providing no evidence of differential pre-recession labor market trajectories.

5.2 Earnings and Employment losses

Baseline results. Table 1 reports estimates of equation (2). Controlling only for life-cycle earnings patterns (Column 1), a one-standard-deviation increase in the provincial shock (8.9 percentage points) reduces cumulative earnings by roughly 4.5 months of pre-recession pay over 2007–2012.¹³ Adding job characteristics, occupational skill group, tenure, experience, and contract type, reduces the coefficient to -2.793 (Column 2). The 34 percent attenuation indicates that more-exposed provinces held workers with less transferable job characteristics, so part of the raw gap reflects worker composition.

Table 1: Cumulative Earnings Impact of the Construction Employment Decline, 2007–2012

	OLS				IV
	(1)	(2)	(3)	(4)	(5)
<i>Dependent variable: Normalized cumulative earnings, 2007–2012</i>					
Shock	-4.241^{***} (0.464)	-2.793^{***} (0.327)	-2.067^{***} (0.292)	-2.026^{***} (0.291)	-3.111^{***} (0.781)
Observations	34,000	34,000	34,000	34,000	34,000
Age $\times$ Gender $\times$ Education	✓	✓	✓	✓	✓
Job characteristics		✓	✓	✓	✓
Regional controls			✓	✓	✓

Notes: The shock is defined as the proportional change in the construction sector’s employment share at the province level between 2007 and 2012. Column (4) uses the proportional change in the number of construction workers by province as an alternative measure. Column (5) reports instrumental variables estimates, instrumenting the shock with average annual growth in construction employment over 2000–2006. Job-level controls include occupational skill group, indicators for part-time and fixed-term contracts, earnings quintiles, and tenure, and experience fixed effects. Regional controls include the local unemployment rate, the initial construction employment share, a Bartik measure of demand shifts in non-construction sectors, and the Herfindahl–Hirschman Index (HHI) for non-construction sectors. Standard errors are clustered at the province level. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Column 3 adds province-level labor-market controls and is the preferred specification.

¹³ $4.241 \times 0.089 \approx 0.377$ times pre-recession annual earnings, equivalent to approximately 4.5 months.

Moving from the 25th to the 75th percentile of the exposure distribution reduces cumulative earnings by approximately 21 percent of pre-recession annual earnings.¹⁴ Column 4 shows nearly identical results when exposure is measured in levels rather than shares, confirming the findings are not an artifact of changes in aggregate provincial employment.

The IV estimate of (-3.111) (Column 5) is somewhat larger in absolute value than the corresponding OLS estimate, but the two remain economically similar. This pattern is consistent with attenuation bias in the OLS specification arising from measurement error in the exposure measure. It may also reflect a local average treatment effect if the instrument places greater weight on provinces where pre-recession construction growth translated more strongly into subsequent labor market exposure. Importantly, instrumenting the shock leaves both the sign and magnitude of the estimated effect largely unchanged, reinforcing the conclusion that the baseline results are unlikely to be driven by omitted provincial factors.

Beneath these average effects lies substantial heterogeneity. Workers facing the same local shock experienced very different outcomes. To understand the sources of these losses, I first decompose aggregate earnings effects into their extensive and intensive margins.

Intensive and extensive margins. Table 2 decomposes the cumulative earnings effect into an extensive margin, days employed over 2007–2012 expressed in years (Column 2), and an intensive margin, average monthly earnings conditional on employment, normalized by the 2005–2006 baseline used for cumulative earnings (Column 3).

Reduced employment duration accounts for roughly 73 percent of the total earnings loss,¹⁵ making it the dominant margin: workers in more-exposed provinces accumulate fewer days of work rather than earning substantially lower wages when employed. The conditional-wage effect is small, consistent with Lachowska et al. (2020), who find that nonemployment duration accounts for the bulk of long-term earnings losses among displaced workers in the US.

The remaining 27 percent is consistent with modest wage declines upon reemployment: construction paid relatively high pre-recession wages (Lacuesta et al., 2020), so workers reemployed elsewhere lost some earnings even in comparable positions. That this wage effect is small relative to the employment margin confirms that job-finding difficulty, not reemploy-

¹⁴The interquartile difference in employment share declines is 0.103 (75th percentile: 0.620; 25th: 0.517). Evaluated at the estimated coefficient: $-2.067 \times 0.103 \approx 0.213$ times annual pre-recession earnings.

¹⁵Cumulative normalized earnings can be expressed as $Y_i = E_i \times 12 w_i$, where E_i denotes years of employment and w_i is average monthly earnings divided by average annual pre-displacement earnings. A first-order decomposition of the shock effect yields $\partial Y / \partial S \approx 12 \bar{w} \cdot (\partial E / \partial S) + 12 \bar{E} \cdot (\partial w / \partial S)$, where $\bar{E} = 3.914$ and $\bar{w} = 0.093$ are sample means. Evaluating: the employment channel contributes $12 \times 0.093 \times (-1.627) = -1.806$ and the wage channel contributes $12 \times 3.914 \times (-0.014) = -0.658$, implying shares of 73 and 27 percent, respectively. The linear approximation overshoots the estimated total (-2.067) because it omits the covariance between E_i and w_i .

ment quality, drives the bulk of the loss, and that displaced workers sorted into sectors paying only modestly below construction rather than moving sharply down the occupation-wage distribution.

Because earnings losses are driven largely by non-employment, adjustment hinges on the availability of suitable reemployment opportunities. Workers whose skills align more closely with the sectors present in their local labor market should receive acceptable offers more quickly, spend less time out of work, and experience smaller cumulative losses. The reallocation index developed in Section 4 is designed to capture this variation in reemployment opportunities.

Table 2: Decomposition of Earnings Losses: Employment versus Wages

	(1) Cumulative Earnings	(2) Cumulative Days Employed	(3) Avg. Earnings
Shock	-2.067*** (0.292)	-1.627*** (0.159)	-0.014* (0.008)
Observations	34,000	34,000	34,000
Controls	✓	✓	✓

Notes: All specifications include the preferred set of individual and regional controls from Column (3) of Table 1. Robust standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Dynamic Adjustment and Persistence

The cumulative estimates leave the timing of adjustment open. Year-by-year effects of the construction shock trace the trajectory of earnings and employment losses:

$$y_{it} = \text{Shock}_{r(i)} \cdot \beta_t + X'_i \Delta_t + \varepsilon_{it}. \quad (3)$$

The dependent variables are annual earnings normalized by 2007 earnings and a binary employment indicator, and the pre-recession coefficients provide a direct test of parallel trends.

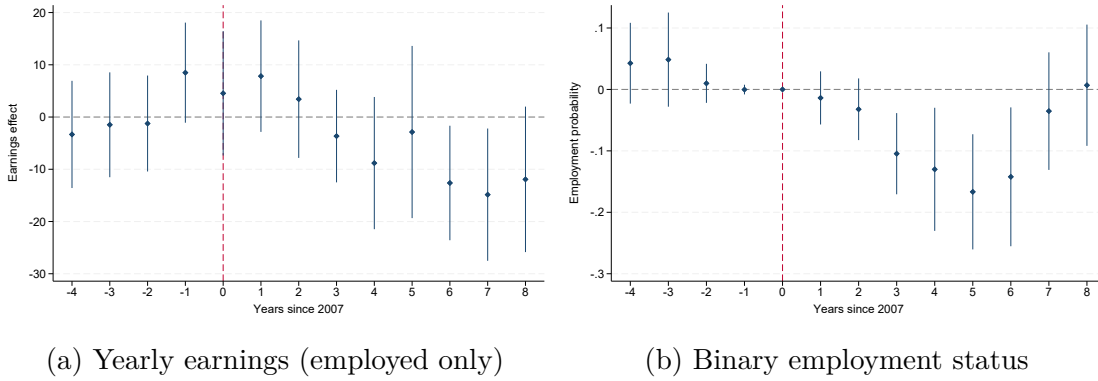
Figure 2 plots the estimated $\hat{\beta}_t$. Pre-recession estimates are small and statistically indistinguishable from zero for both outcomes, supporting the identifying assumption. After 2007, employment probabilities in more-exposed provinces decline sharply; conditional wages remain near pre-recession levels throughout while employment probabilities stay depressed for several years, reinforcing the decomposition's finding that non-employment, not wage compression, drives cumulative losses.

At the 2012 trough, a one-standard-deviation larger shock lowers the annual employment probability by roughly 1.5 percentage points.¹⁶ The gap narrows rapidly: the coefficient falls to -0.035 by 2014, a 77 percent compression, and reaches $+0.007$ by 2015, statistically indistinguishable from zero, so workers across the exposure distribution converge to similar annual employment rates within eight years.

The convergence of annual employment rates, therefore, coexists with substantial differences in cumulative employment histories. Although employment probabilities are similar by 2015, workers in more exposed provinces have spent considerably more time out of work, generating losses that subsequent recovery cannot fully reverse. Since construction itself did not rebound, these employment gains reflect reallocation into other sectors rather than re-entry into the original industry. Section 6 documents this adjustment process and shows that migration plays little role.

Combined with the absence of meaningful wage effects, these results indicate that long-run recovery depends primarily on how quickly workers reenter employment. The next results examine which worker characteristics are associated with faster adjustment and uses this evidence to motivate a measure of reallocation capacity.

Figure 2: Year-by-Year Impact of the Construction Employment Decline



Notes: Coefficients from equation (3). Additional controls as in Column (3) of Table 1. Confidence intervals based on robust standard errors clustered at the province level.

Heterogeneity. The average effects mask substantial heterogeneity. To assess which workers are most affected, I interact the provincial shock with baseline worker characteristics (Table A2). Workers with less labor-market experience, lower pre-recession earnings, and younger age profiles experience significantly larger cumulative earnings losses following the collapse. The estimated effects are also larger for workers with lower education and shorter tenure, although these differences are less precisely estimated. Occupational skill group is

¹⁶ $0.167 \times 0.089 \approx 0.0149$. At the interquartile difference (10.3 percentage points), the implied employment probability gap is $0.167 \times 0.103 \approx 0.017$ in 2012.

the notable exception: once other characteristics are controlled for, low-skilled workers do not experience significantly larger losses than their high-skilled counterparts.

Figure A16 documents the magnitude of these differences. Workers younger than 30 at the recession’s onset experienced the largest losses of any age group, consistent with their disproportionate concentration in temporary positions with limited transferability outside construction. Less-experienced workers and those in the lower half of the pre-recession earnings distribution likewise suffered substantially larger losses. The skill-group null sharpens the picture: once earnings and experience are controlled for, occupational classification adds little, suggesting that accumulated human capital rather than occupational category governs reallocation prospects. These patterns motivate but do not identify the mechanism. Observable characteristics leave substantial residual heterogeneity unexplained: within any age, education, or earnings cell, workers in the same province face similar shock exposure yet experience markedly different adjustment trajectories. Section 6 uses the reallocation index to explain this within-cell variation.

6 Skill–Demand Alignment and Adjustment Heterogeneity

Earnings losses are driven primarily by prolonged non-employment rather than by wage penalties conditional on reemployment (Section 5), so the speed of reentry governs the bulk of post-displacement costs. Whether that speed depends on the alignment between worker skills and locally available non-construction sectors, captured by the reallocation index, is the question this section takes up: if alignment shortens non-employment, workers facing less overlap with local sectors should sustain larger cumulative losses regardless of aggregate provincial conditions.

6.1 Reallocation Capacity and Earnings Losses

I augment equation (2) with the reallocation index:

$$y_i = \theta \text{Shock}_{r(i)} + \gamma \text{RI}_i + \delta \text{Shock}_{r(i)} \times \text{RI}_i + X_i' \beta + \varepsilon_i \quad (4)$$

where RI_i is the reallocation index constructed in Section 4, which varies across workers within the same province, and $\text{Shock}_{r(i)}$ as in equation (2).

Consistent with the predictions of Section 4, the reallocation index is positively associated with cumulative days employed (Column 1 of Table 3), implying that workers with

Table 3: Labor Market Impact of the Construction Employment Decline, 2007–2012

	Employment		Cumulative Earnings		Avg. Earnings	
	(1)	(2)	(3)	(4)	(5)	(6)
Shock	−1.617*** (0.156)	−1.616*** (0.156)	−2.045*** (0.286)	−2.045*** (0.278)	−0.013* (0.007)	−0.013* (0.007)
Reall. Index	0.045** (0.020)	−0.149*** (0.051)	0.096*** (0.028)	−0.105 (0.098)	0.001** (0.000)	0.000 (0.002)
Shock × Reall. Index		0.332*** (0.080)		0.344** (0.162)		0.002 (0.003)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are cumulative days employed annualized (Columns 1–2), normalized cumulative earnings (Columns 3–4), and average monthly earnings conditional on employment (Columns 5–6), each measured over 2007–2012 and normalized by the worker’s average annual earnings in 2005–2006. The shock measures the proportional change in the construction sector’s employment share at the province level between 2007 and 2012. The reallocation index, constructed using pre-recession data, measures the alignment between each worker’s skill profile and the sectoral composition of local labor demand outside construction. Odd-numbered columns include the reallocation index as a main effect; even-numbered columns add its interaction with the shock. All specifications include worker, job, and regional controls from Column 3 of Table 1. Standard errors are clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

stronger reallocation prospects return to work more quickly. The key coefficient is the interaction between the index and the shock, which captures whether worker-specific reallocation opportunities mitigate the effects of a given provincial contraction. At the 75th percentile of the provincial shock distribution, a one-standard-deviation improvement in the index attenuates the cumulative employment loss by 20.6 percent of the shock-induced employment penalty.¹⁷

The interaction is also positive and significant for cumulative earnings (0.344, Column 4): at a given level of exposure, the corresponding increase in the index offsets approximately 16.8 percent of the earnings loss.¹⁸

The underlying margin is unambiguous. The employment interaction equals 0.332 (0.080), whereas the conditional-wage interaction is 0.002 (0.003), effectively zero both statistically and economically. This divergence is difficult to reconcile with a match-quality mechanism in which high-index workers access intrinsically better-paying sectors, since such a mechanism would generate wage gains alongside employment gains. Instead, the evidence points to a search-based channel. The index mitigates earnings losses by accelerating reemployment,

¹⁷The employment penalty at the 75th percentile of the shock is $-1.616 \times 0.620 \approx -1.002$; the attenuation from a one-standard-deviation improvement in the index is $0.332 \times 0.620 \approx 0.206$.

¹⁸The earnings penalty at the 75th percentile of the shock is $-2.045 \times 0.620 \approx -1.268$; the attenuation is $0.344 \times 0.620 \approx 0.213$, yielding $0.213/1.268 \approx 16.8$ percent.

not by raising wages conditional on reemployment. It therefore captures access to suitable employment opportunities rather than the quality of the eventual match, consistent with the interpretation developed in Section 4. This is consistent with evidence that displaced workers, particularly during large recessions, accept reemployment in lower-paying occupations rather than waiting for quality matches (Huckfeldt, 2022), reinforcing that the binding constraint is access to any suitable job, not the wage premium of the eventual one.

To assess whether the moderating role of reallocation capacity varies systematically across the index distribution, I interact the shock with quartile indicators:

$$y_i = \sum_{k=1}^4 \beta_k Q_i^k \cdot \text{Shock}_{r(i)} + X_i' \Delta + \varepsilon_i, \quad (5)$$

where Q_i^k indicates that worker i belongs to quartile k of the index distribution. If reallocation capacity governs adjustment, shock coefficients should decline across quartiles, with better-aligned workers losing less under the same shock.

Table 4: Sectoral Composition and the Consequences of the Construction Collapse: Heterogeneity by Reallocation Index Quartile

	Employment		Cumulative Earnings		Avg. Earnings	
	(1)	(2)	(3)	(4)	(5)	(6)
Shock	-1.627*** (0.159)		-2.067*** (0.292)		-0.014* (0.008)	
<i>Shock × Reallocation Index Quartile:</i>						
Q1 (lowest alignment)		-1.932*** (0.237)		-2.462*** (0.369)		-0.021** (0.010)
Q2		-1.710*** (0.215)		-2.039*** (0.368)		-0.000 (0.014)
Q3		-1.504*** (0.257)		-2.022*** (0.381)		-0.016* (0.009)
Q4 (highest alignment)		-1.268*** (0.235)		-1.590*** (0.507)		-0.011 (0.013)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are cumulative days employed annualized (Columns 1–2), normalized cumulative earnings (Columns 3–4), and average monthly earnings conditional on employment (Columns 5–6), each measured over 2007–2012 expressed relative to the worker’s baseline average annual earnings in 2005–2006. Columns (1), (3), and (5) reproduce the baseline shock coefficient from Table 1. Columns (2), (4), and (6) introduce interactions between the shock and quartile indicators for the reallocation index, allowing the shock coefficient to vary across the index distribution. All specifications include worker, job, and regional controls from Column (3) of Table 1. Standard errors are clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The employment gradient is monotonic and precisely estimated (Table 4). Workers in the highest quartile of the reallocation index are substantially less affected by the construction shock than those in the lowest quartile. The estimated employment penalty declines by 34 percent across quartiles, from -1.932 in Q1 to -1.268 in Q4. An F -test rejects equality of the Q1 and Q4 coefficients ($p < 0.05$), confirming that workers in the highest reallocation quartile experience significantly smaller employment losses than observationally similar workers in the lowest quartile. The gradient is concentrated at the tails of the distribution: the Q1–Q2 difference (0.222) and the Q3–Q4 difference (0.236) both exceed the Q2–Q3 difference (0.206). At the 75th percentile of the provincial shock distribution, these estimates imply a loss of 1.198 years of employment for a worker in the bottom quartile of the index, compared with 0.786 years for an otherwise identical worker in the top quartile—a difference of 0.412 years, or nearly five months.¹⁹

The cumulative earnings gradient mirrors the employment results but is estimated less precisely. The earnings penalty falls by 35 percent across quartiles, from -2.462 in Q1 to -1.590 in Q4. By contrast, conditional wages exhibit no monotonic relationship with the reallocation index. This pattern accords with the interpretation developed in Section 4: the index captures access to alternative employment opportunities through worker–demand alignment rather than wage-setting conditional on reemployment. Workers in better-matched local labor markets return to work more quickly and spend more time employed, but do not systematically obtain higher wages once reemployed. Differences in search frictions, monopsony power, or sector-specific wage premia may influence accepted wages without materially affecting the speed or likelihood of reallocation.

This quartile spread exceeds the variation associated with any single observable in Section 5.3, including the education and age-below-30 contrasts.²⁰

The evidence indicates that the Reallocation Index operates primarily through reemployment rates rather than through the wages of subsequent matches. The interaction between the shock and the index is large for employment outcomes (0.346) but close to zero for wages (0.001). Consistent with this pattern, employment effects increase monotonically across index quartiles and are estimated precisely, whereas wage differences are neither monotonic nor statistically distinguishable from zero. The same ordering emerges in the inference exercises: employment effects remain highly significant under the wild bootstrap ($p = 0.000$), earnings effects are weaker ($p = 0.071$), and wage effects are statistically insignificant throughout. Taken together, these results suggest that the index captures variation in the rate at which

¹⁹Penalties at the 75th percentile of the shock are $-1.932 \times 0.620 \approx -1.198$ years (Q1) and $-1.268 \times 0.620 \approx -0.786$ years (Q4).

²⁰The comparison is approximate: the subgroup regressions in Table A2 use split samples, whereas the quartile interactions use the full sample.

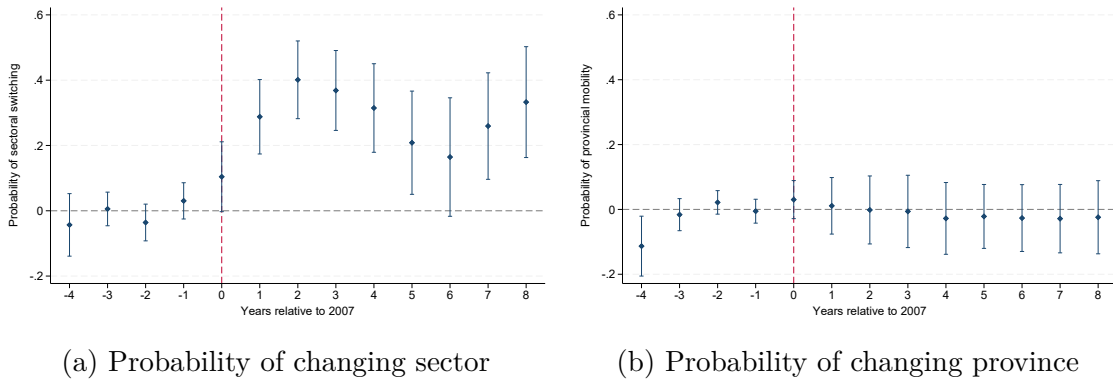
displaced workers receive acceptable job offers from surviving sectors. By contrast, explanations based on systematic differences in match quality, access to higher-paying industries, or the capture of sector-specific wage premia receive little support.

Demographic characteristics capture mean differences across broad groups but leave substantial within-group variation: two workers identical in age, education, and pre-recession earnings face different prospects when local sectoral structure suits one better, and the index isolates this residual component that demographics cannot measure. Whether this within-province, within-shock heterogeneity translates into faster sectoral reentry among higher-index workers is examined in Section 6.2.

6.2 Sectoral Reallocation Under Geographic Immobility

Displaced workers can adjust along two margins: moving to another province or leaving construction for another sector. Replacing cumulative earnings in equation (2) with a linear probability model for each, Figure 3 plots year-by-year shock coefficients over 2003–2015.²¹

Figure 3: Probability of Sectoral versus Geographic Reallocation



Notes: Panel (a) reports the probability of transitioning out of the construction sector into a different sector; Panel (b) reports the probability of residing in a province different from the 2007 province of residence. The sample and control variables follow the baseline specification in Table 1. Both panels include the pre-period (2004–2006) average of the relevant mobility indicator as a control.

The two margins diverge sharply: sectoral transitions respond strongly and persistently while geographic mobility does not. Pre-recession sectoral coefficients (2003–2005) are near zero (Panel a), confirming parallel pre-shock trajectories. The sectoral response emerges in 2008 and peaks in 2009, when the interquartile shock difference raises the annual transition probability by roughly 5.0 percentage points, and coefficients stay large and precise through

²¹Figure A18 presents the same analysis without conditioning on pre-recession average mobility. The results are qualitatively unchanged, although the estimates are somewhat less precise.

2015 (3.8 to 4.2 points at the interquartile difference), reflecting continued reallocation out of a sector whose employment share remained permanently depressed.

Geographic mobility does not respond in economically or statistically meaningful ways over 2007–2015 (Panel b). Point estimates of the interquartile effect never exceed 0.5 percentage points, corresponding to less than 4 percent of the roughly 12 percent pre-recession baseline migration rate, and are statistically indistinguishable from zero throughout the post-recession period. Pre-trends are flat: coefficients for 2004–2006 are small and not statistically different from zero, consistent with parallel trends in geographic mobility. The 2003 coefficient is negative and marginally significant (-0.091 , $p < 0.10$), suggesting slightly lower pre-recession mobility in more-exposed provinces, although the magnitude is small relative to post-recession sectoral adjustments.

A potential concern is that correlated shocks across connected labor markets may attenuate measured out-migration by depressing employment opportunities in destination provinces. To address this, I construct a destination-adjusted exposure measure that nets out such correlated demand shocks using pre-recession bilateral migration flows (Bertoli and Moraga, 2013; Borusyak et al., 2022).

$$\text{Shock}_r^{\text{adj}} = \text{Shock}_r - \sum_{k \neq r} \mu_{r \rightarrow k} \text{Shock}_k. \quad (6)$$

Table 5 quantifies both margins at the cross-sectional level. At the interquartile shock difference, sectoral exit rises by roughly 3.9 percentage points (Columns 4–6) while inter-provincial migration is unchanged, under 0.3 percentage points or 6 percent of the pre-recession baseline (Columns 1–3). Both results hold under the destination-adjusted shock of equation (6) (Columns 3 and 6), giving a sectoral-to-geographic adjustment ratio of roughly 13 to 1.

If adjustment runs mainly through sectoral rather than geographic reallocation, the index should strongly capture who exits construction: the shock should accelerate departure for workers whose skills map onto available local demand and leave exit rates unchanged for those with weaker alignment. Table 6 confirms it: the shock raises the exit probability by $0.520 \times \text{Shock}$ for top-quartile workers against $0.283 \times \text{Shock}$ at the bottom; at the mean Q1 shock of 0.574, this is roughly 30 versus 16 percentage point.²² Workers with the weakest skill–demand alignment show no greater propensity to leave construction as local conditions deteriorate, and these are the same workers who bear the largest and most persistent non-

²²Table A7 in the Appendix uses as the dependent variable the probability of employment outside construction in 2012, showing an even more striking difference between workers in the upper quartiles of the reallocation index and those at the bottom.

Table 5: Geographic versus Sectoral Reallocation

	(1)	(2)	(3)	(4)	(5)	(6)
	Changed province			Changed sector		
Shock	−0.006 (0.120)	0.057 (0.091)		0.374*** (0.055)	0.377*** (0.054)	
Weighted shock			−0.047 (0.166)			0.288*** (0.079)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: The dependent variable in Columns (1)–(3) is an indicator for residing in a province different from the 2007 province of residence, measured in 2012. Columns (4)–(6) use an indicator for employment outside construction in 2012. Columns (2) and (5) additionally control for pre-recession average mobility, and Columns (3) and (6) control for the pre-migration adjusted shock. The adjusted shock nets out construction employment declines in connected provinces, weighted by pre-recession bilateral migration flows following equation (6). All specifications include the preferred set of individual and regional controls from Column (3) of Table 1. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

employment costs of Section 6.1.

Figure A2 traces these dynamics. Pre-recession trajectories are parallel across quartiles, supporting the identifying assumption of no differential pre-trends. Following the shock, workers in the top quartile attain substantially higher and more persistent non-construction employment than those in the bottom two quartiles, with little evidence of convergence over the subsequent seven years. Bottom-quartile workers exhibit a brief employment advantage in 2008, but this reverses by 2009 and widens thereafter. Reallocation destinations also differ systematically in the quality of jobs (Table A11). Bottom-quartile workers who leave construction disproportionately sort into hospitality and agriculture—cyclical sectors characterized by less stable employment—whereas top-quartile workers reallocate into sectors with more durable employment opportunities. The combination of persistent employment gaps and systematically weaker destinations points to a binding reallocation constraint. For low-index workers, limited occupational overlap with local labor demand restricts access to viable alternatives, leaving the extensive margin of adjustment largely unavailable.

6.3 Index Decomposition

Whether skill alignment governs the magnitude of earnings losses depends on the underlying channel: the match between worker characteristics and local sectoral composition, the wage premia of locally available sectors, or a combination of both. I distinguish these mechanisms using three measures. The baseline reallocation index weights skill–demand alignment by

Table 6: Sectoral Composition and the Probability of Leaving Construction

	(1)	(2)	(3)
	Sector Change, 2008–2012		
Shock	0.374*** (0.055)	0.376*** (0.051)	
Reall. Index		−0.032** (0.015)	
Shock × Reall. Index		0.074*** (0.024)	
<i>Shock × Reallocation Index Quartile:</i>			
Q1 (lowest)			0.283*** (0.059)
Q2			0.282*** (0.071)
Q3			0.413*** (0.071)
Q4 (highest)			0.520*** (0.056)
Controls	34,000 ✓	34,000 ✓	34,000 ✓

Notes: Linear probability estimates. Dependent variable: indicator for employment in a sector other than construction at any point between 2008 and 2012. All specifications include the preferred set of individual and regional controls from Column (3) of Table 1. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

each sector’s median wage relative to construction. Removing the wage weights yields a pure skill–sector overlap measure that isolates the job-matching component. A third measure captures the local sectoral wage mix, constructed independently of skill alignment, and isolates whether a favorable wage distribution alone can account for the results. All three variables are standardized, so coefficients are expressed as the effect of a one-standard-deviation change.

The no-wage variant of the index retains strong predictive power (0.307), while the wage-mix variant flips sign (−0.482 in Column 2, −1.121 in Column 3), indicating that the predictive content derives from alignment between worker skills and the composition of surviving local demand rather than from differential local access to high-wage sectors (Table 7). Workers in provinces with higher-paying surviving industries do not recover faster; if anything, conditional on skill alignment, they fare worse. The binding constraint on post-displacement earnings recovery is occupational mismatch, not the wage level of available destinations. This finding echoes Autor and Dorn (2013) and Lise and Postel-Vinay (2020), who document persistent reallocation penalties for workers whose skills transfer poorly across sectors.

Match is non-separable across levels: a worker’s alignment reflects both the opportunity set common to their province and their position relative to observationally similar workers within it. Treating the index as a single regional component might leave the second source of variation unexploited. I therefore decompose it into a province mean, capturing the average local opportunity set, and a within-province residual, capturing alignment differences among

Table 7: Reallocation Index Variants

	Employment			Cumulative Earnings		
	Baseline (1)	No Wage (2)	Wage-mix (3)	Baseline (4)	No Wage (5)	Wage-mix (6)
Shock	-1.617*** (0.156)	-1.656*** (0.165)	-1.645*** (0.143)	-2.045*** (0.277)	-2.115*** (0.294)	-2.172*** (0.271)
Shock $\times$ Index	0.347*** (0.083)	0.307*** (0.073)	-0.482*** (0.155)	0.353** (0.160)	0.225 (0.155)	-1.121*** (0.364)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Each column reports OLS estimates of the construction shock interacted with one variant of the reallocation index. Columns (1) and (4) use the baseline index $\sum_j \hat{\phi}_j(X_i) \times \hat{s}_{jr}/\bar{s}_j \times \widetilde{W}_j$, which combines skill-demand alignment, local vacancy intensity, and the relative wage weight. Columns (2) and (5) drop the relative wage ratio $\widetilde{W}_j$, isolating the skill-demand alignment component. Columns (3) and (6) replace worker-specific transition probabilities $\hat{\phi}_j(X_i)$ with uniform weights $1/J$, so that the index captures only the local sectoral wage mix independently of skill alignment. All index variants are standardised to mean zero and unit standard deviation. Level terms for each index are included in all specifications but not reported. All regressions include the individual controls in the baseline specification of Table 1. Standard errors clustered at the province level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

observationally similar workers in the same province.

Table 8 reports the two components separately and jointly. For cumulative days employed the within-province component dominates: in the joint specification the shock interaction with the worker-level component is 0.296, roughly 2.2 times the regional component (0.137). The regional interaction loses precision when both enter jointly, but its magnitude, nearly half the within-province estimate, indicates that provincial structure contributes to employment recovery even where the available variation cannot identify it separately. Reemployment speed thus depends primarily on how well a worker’s skills match locally available sectors, with provincial structure secondary but non-negligible.

The pattern reverses for cumulative earnings, with a near-identical ratio in the opposite direction. Provincial structure, through the sectoral presence, size, and wages a worker can land in, therefore accounts for most of the shock-driven variation in earnings losses, with worker-level alignment accounting for the remainder. This parallels Yi et al. (2024), who show that local industry composition substantially moderates earnings losses from trade shocks; the decomposition here complements that evidence by recovering the worker-level alignment component within the region-industry margin.

The 2:1 ratio across outcomes is informative about the underlying mechanism. Employment outcomes are governed by worker-level occupational compatibility, while earnings additionally reflect differences in sectoral wage premia across destinations. These distinct

Table 8: Regional versus Worker-Level Reallocation Index

	Employment			Cumulative Earnings		
	(1) Regional	(2) Worker	(3) Joint	(4) Regional	(5) Worker	(6) Joint
Shock	−1.569*** (0.145)	−1.616*** (0.156)	−1.569*** (0.145)	−1.976*** (0.247)	−2.045*** (0.278)	−1.977*** (0.249)
Shock × Regional	0.144 (0.133)		0.137 (0.134)	0.457* (0.233)		0.441* (0.234)
Shock × Worker-level		0.332*** (0.080)			0.344** (0.162)	
Shock × Within			0.296*** (0.063)			0.201 (0.135)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are cumulative years of employment (columns 1–3) and cumulative normalized earnings (columns 4–6), 2007–2012. The reallocation index measures the expected value of transitioning out of construction, standardized to mean zero and unit variance. The regional index is the province mean of the reallocation index, standardized to unit variance; the within-province index is the individual deviation from the province mean, standardized to unit variance. Both components enter columns (3) and (6) simultaneously. The regional and within-province components are constructed as orthogonal on the raw scale. All specifications include individual, contract, and regional controls. Clustered standard errors at the province level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

margins imply that reallocation capacity operates mainly through the speed of reemployment, with limited effects on post-reemployment wages.

Place-based strategies targeting high-wage industries rely on agglomeration-driven spillovers in employment (Moretti, 2011), while diversified local economies are expected to facilitate adjustment through multiple reallocation channels (Jacobs, 1984). The evidence here suggests these mechanisms are strongly conditional on worker–sector compatibility. Reallocation occurs only when worker skills align with local labor demand, and adjustment costs increase with occupational distance (Gathmann and Schönberg, 2010). During the Great Recession, displaced construction workers did not move systematically into expanding sectors such as finance, healthcare, or professional services because occupational mismatch constrained feasible transitions. This implies that agglomeration and diversity are not sufficient for adjustment; the relevant margin is the ability to traverse occupational boundaries.

7 Policy Counterfactual

Improving a worker’s index position is analogous to shifting the sector-matching function upward—the channel through which occupational retraining and targeted job-search assistance operate. Using the quartile coefficients of Table 4, this section asks how cumulative earn-

ings would change if active labor market programs improved the reallocation capacity of the worst-aligned workers. The exercise is partial-equilibrium: it uses reduced-form estimates to bound earnings gains, not to identify net welfare effects.

The calculation abstracts from general equilibrium feedback: it holds fixed the provincial shock, the post-recession demand composition, and untreated workers' outcomes, and assumes shifting workers toward higher-index states does not crowd out others or compress wages in receiving sectors. The exercise applies to programs that accelerate sectoral reentry—retraining and targeted job-search assistance—not to income support, which does not move workers along the index distribution. Because program costs of shifting a worker across quartiles are not identified, the estimated gains are earnings bounds rather than net welfare effects.

Spanish ALMP spending during the Great Recession was not systematically conditioned on workers' reallocation capacity. Given the gap between bottom- and second-quartile shock coefficients (2.462 versus 1.590), a uniform allocation rule leaves predictable heterogeneity in losses unaddressed.

Define the actual cumulative earnings loss for worker i in quartile k as $L_i^{\text{actual}} = \hat{\beta}_k \times \text{Shock}_{r_i}$. Under the targeted counterfactual, treated Q_1 workers experience $L_i^{\text{CF1}} = \hat{\beta}_{Q_2} \times \text{Shock}_{r_i} = -2.039 \times \text{Shock}_{r_i}$, while workers in Q_2 through Q_4 remain at actual outcomes. The aggregate per-worker recovery, averaged over the full sample, is

$$\Delta^{\text{CF1}} = \frac{N_{Q_1}}{N} \cdot (\hat{\beta}_{Q_1} - \hat{\beta}_{Q_2}) \cdot \overline{\text{Shock}}, \quad (7)$$

where $\overline{\text{Shock}} = 0.574$ is the mean shock among Q_1 workers and $N_{Q_1}/N = 0.25$ by construction.

Moving Q_1 workers to Q_2 alignment recovers roughly 17 percent of the treated group's earnings loss; scaled by the Q_1 share, the aggregate per-worker recovery is approximately 5 percent of the full-sample average loss.²³

A more intensive scenario moves Q_1 workers to the Q_4 distribution, approximating a comprehensive skill-reorientation program. Because the recovery and the baseline loss scale with the same shock, the recovered share is invariant to the shock level: the shift recovers about a third of the treated group's loss, or roughly 10 percent of aggregate sample losses once weighted by the Q_1 share.²⁴

²³Per-unit-shock recovery: $\hat{\beta}_{Q_1} - \hat{\beta}_{Q_2} = 0.423$. Per-treated-worker recovery: $0.423 \times 0.574 \approx 0.243$. Aggregate per-worker recovery: $0.243 \times 0.25 \approx 0.061$, against a full-sample average loss of $2.067 \times 0.574 \approx 1.186$, giving 5.1 percent. The treated group's baseline loss is $2.462 \times 0.574 \approx 1.413$, so the shift recovers $0.423/2.462 \approx 17$ percent of it.

²⁴Per-unit-shock recovery: $\hat{\beta}_{Q_1} - \hat{\beta}_{Q_4} = 0.872$, so the treated share is $0.872/2.462 \approx 0.35$ and the aggregate

The intensive-scenario gain—recovering roughly a third of the treated group’s loss—is non-negligible against typical European ALMP returns (Card et al., 2018), though closing the entire Q_1 -to- Q_4 gap still leaves most of the bottom-quartile loss intact. Targeting is feasible ex ante: the index is constructible from pre-displacement administrative records on education, occupational skill group, and province, and captures adjustment heterogeneity not explained by any single observable characteristic (Table A2).

8 Additional Robustness Checks

Sample composition. The baseline results are insensitive to the age ceiling and the restriction to native workers (Table A9). Extending the sample to workers aged up to 55 and additionally incorporating foreign-born workers leaves both the earnings and employment coefficients within five percent of the baseline estimates, despite immigrants’ well-documented tendency to respond to local demand shocks through geographic relocation (Cadena and Kovak, 2016).

Minimum attachment threshold. The earnings and employment coefficients are stable across minimum-attachment thresholds of three, six, nine, and twelve months, spanning a 22 percent variation in sample size (Table A10). The stability at the three-month threshold—which admits workers with substantially weaker pre-recession attachment—addresses the concern that stricter cutoffs select on workers whose 2007 employment was itself a product of the boom.

Instrumenting the shock. Instrumenting the provincial shock and its interaction with the index leaves the central result intact and strengthens it: the shock–index interaction rises from 0.332 to 0.570 for employment and from 0.344 to 0.703 for earnings (Table A5). Two forces likely account for the amplification. Classical measurement error in the provincial shock attenuates the OLS interaction toward zero. The IV estimates also reflect a local average treatment effect concentrated in provinces where pre-recession construction growth was most pronounced, which tend to be those where the collapse was largest. At a 75th-percentile shock, a one-standard-deviation improvement in alignment offsets about one-fifth of losses under OLS and one-third under IV. First-stage diagnostics support this reading: Sanderson-Windmeijer conditional F-statistics are 12.51 for the shock and 39.96 for the interaction, and the Anderson–Rubin test rejects jointly zero coefficients ($F = 11.84$, Table A6).

Inference robustness. Table A8 reports wild bootstrap p-values from 9,999 replications with Rademacher weights, clustering by province. The shock is significant at the one percent level for cumulative employment and earnings; the p-value of 0.063 for average wage reflects

share, weighted by $N_{Q1}/N = 0.25$, is approximately 10 percent.

the noisier character of this outcome throughout the analysis. The $\text{Shock} \times RI^w$ interaction is significant at the one percent level for cumulative employment ($p = 0.000$) and at the ten percent level for cumulative earnings ($p = 0.071$), and is not significant for average wage. The finite-sample correction modestly widens inference for the earnings interaction while leaving the employment result unchanged.

Province fixed effects. Table A13 adds province fixed effects, absorbing the shock’s main effect and all time-invariant provincial confounds, to confirm that the $\text{Shock} \times RI^w$ interaction is identified from within-province cross-worker variation. The employment interaction is essentially unchanged at 0.322 (SE = 0.070, $t = 4.60$), a 3 percent attenuation relative to the baseline 0.332. The earnings interaction attenuates to 0.219 and loses precision, consistent with the decomposition in Table 8: cumulative earnings reflect both reemployment speed—a within-province worker-level object—and the sectoral wage structure of reemployment destinations, which varies primarily across rather than within provinces. Since the model’s central prediction concerns offer arrival rates, the employment result is the primary mechanism test, and it survives.

Reallocation index and worker and regional characteristics. A potential concern is that the reallocation index captures variation in observable worker characteristics—education, occupational skill, labor market experience—or provincial conditions rather than skill–demand alignment per se. Table A12 addresses this by augmenting the baseline interaction with shock interactions for each characteristic separately. The $\text{Shock} \times RI^w$ coefficient is stable across columns (1)–(6): it ranges from 0.267 to 0.332, a maximum attenuation of 20 percent relative to the baseline, and remains significant at conventional levels throughout. The HHI and unemployment rate interactions (columns 5–6) leave the coefficient essentially unchanged at 0.308, ruling out the interpretation that the index proxies for local labor market diversity or pre-recession slack. A13

Reallocation dynamics. Year-by-year estimates of the shock interacted with bottom- and top-quartile index indicators show no differential pre-trends: Q1 and Q4 coefficients are small and statistically indistinguishable from zero throughout 2003–2007 on both employment and sectoral exit, so the index does not proxy pre-existing differences in worker outcomes (Figure A2). The Q1–Q4 gap opens immediately after the shock, deepens through $t = 3$ – 4 , and persists through $t = 8$ without convergence, ruling out a transitory interpretation of the moderating effect.

Reemployment hazard. Table A14 estimates cause-specific reemployment hazards separately for workers who transition immediately following displacement and for those who experience a non-employment spell, providing a direct test of Proposition 1(i). Among workers who move directly into another job, the hazard ratio for the highest alignment quartile is

1.906 relative to otherwise similar workers exposed to the same shock. Among workers who enter non-employment, the corresponding hazard ratio is 3.286.²⁵ Both estimates support the proposition that better-aligned workers reallocate more rapidly following displacement.

9 Conclusion

This paper studies how local sectoral composition shapes worker adjustment to large, persistent sectoral shocks. Using Spain’s construction collapse—a contraction of roughly 60 percent that was spatially uneven—I document three findings consistent with the alignment mechanism. First, displaced workers suffered large and persistent earnings losses operating predominantly through extended non-employment: roughly three-quarters of the cumulative loss reflects reduced time in employment rather than lower wages, which remain near pre-recession levels throughout. Second, adjustment ran through intersectoral mobility rather than geographic relocation—the sectoral transition response is roughly thirteen times the geographic response across the full post-recession period. Third, and most centrally, workers whose skill profiles aligned with locally available sectors experienced substantially smaller earnings losses and shorter non-employment spells.

The paper’s central contribution is a worker-level reallocation index that formalizes this mechanism as an offer-arrival rate: it predicts how fast a worker exits non-employment and how readily she switches sectors, but not the conditional wage she commands once matched. The employment margin responds sharply to the index while conditional wages show no meaningful response—the pattern an offer-arrival account predicts, and the opposite of what access to structurally better-paying sectors would imply. This partition is what separates the alignment mechanism from match-quality and wage-setting alternatives.

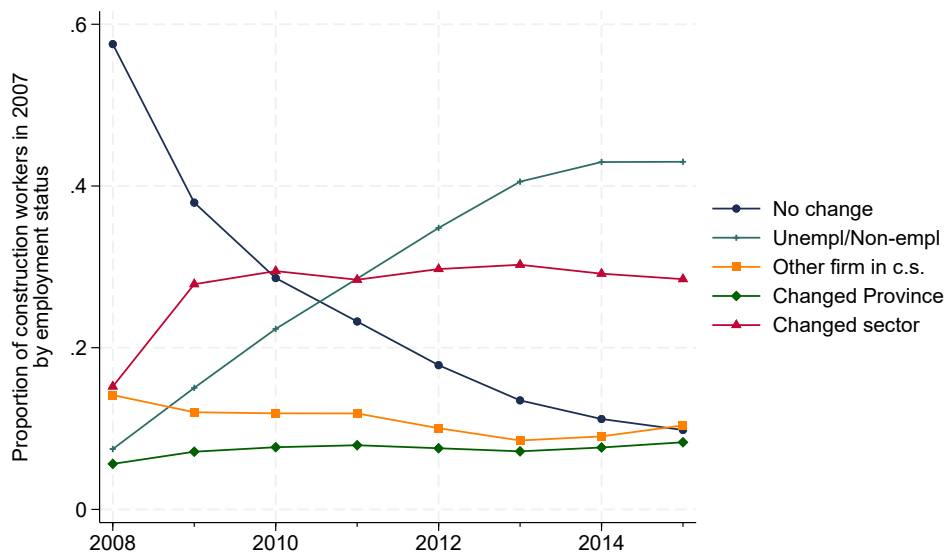
The mechanism is not specific to Spain. Its logic requires, most importantly, that geographic mobility be incomplete in the short and medium run and that local economies differ in sectoral composition—conditions that characterize many episodes of structural change, including trade-exposed manufacturing regions (Autor et al., 2014; Dix-Carneiro and Kovak, 2017) and labor markets reshaped by automation (Acemoglu and Restrepo, 2020). The index is constructible from matched worker–firm data and a pre-shock window, which are increasingly available across OECD countries, so both the object measured and the identification conditions are portable. The estimated magnitudes are specific to Spain; the structure is not.

²⁵The Q1–Q4 contrast is estimated imprecisely among non-employment spells, reflecting the relatively small number of observed failures in this subsample, but point estimates are consistently positive and economically large across specifications.

The results carry direct implications for adjustment policy. Standard programs—Trade Adjustment Assistance, regional retraining schemes, sectoral transition funds—implicitly treat all workers within a location as facing identical adjustment constraints. This paper shows that such targeting is misspecified: within-market heterogeneity in reallocation capacity is large and predictable *ex ante* from administrative records. Active labor market programs are likely to generate larger returns when they direct training and job-search assistance toward workers whose skill profiles are poorly aligned with locally growing sectors, rather than toward nationally high-return skills with limited local demand. The equilibrium effects of such interventions on local sectoral composition—and hence on future reallocation capacity—remain an open question, and one with direct bearing on whether place-based diversification can reduce the adjustment costs documented here.

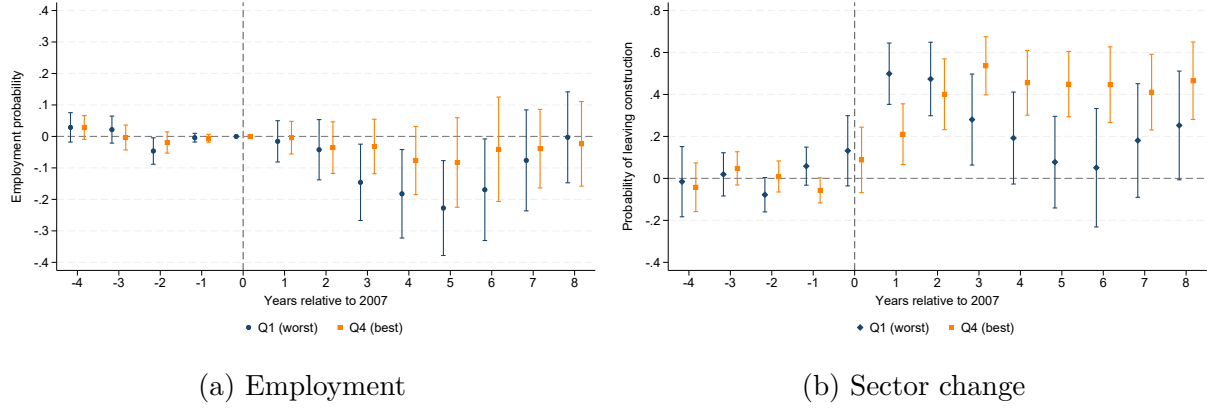
A Additional Figures

Figure A1: Working status of individuals employed in the construction sector in 2007



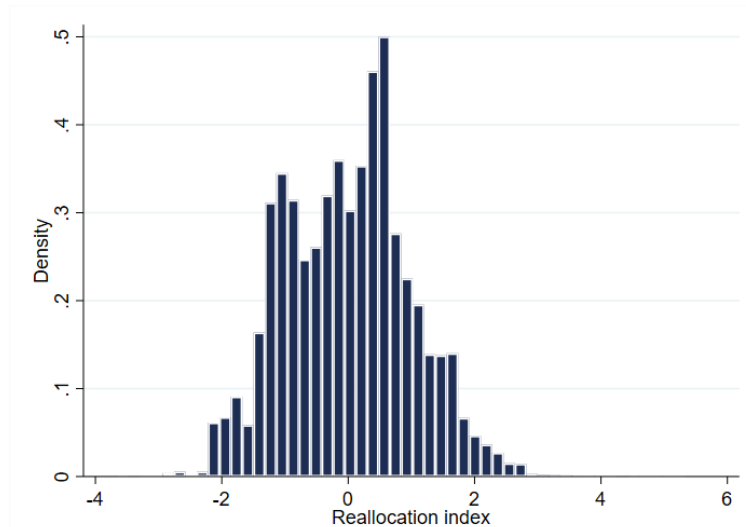
Notes: The sample includes native workers employed in the construction sector in 2007. Employment status is tracked annually through 2015.

Figure A2: Probability of Employment and Sectoral Exit by Reallocation Index Quartile



Notes: Year-by-year linear probability estimates. Dependent variable: indicator for employment and for employment in a sector other than construction in year t , for $t \in \{2003, \dots, 2015\}$. Each year is estimated separately as a regression of the annual sector-change indicator on the shock interacted with indicators for each quartile of the reallocation index, conditional on quartile fixed effects and the pre-shock (2003–2006) average of the sector-change indicator. Controls and sample restrictions follow Table 1. Standard errors clustered at the province level.

Figure A3: Distribution of the Reallocation Index



Notes: Distribution of the normalized reallocation index among workers employed in the construction sector in 2007. The measure is standardized to have mean zero and unit standard deviation in the estimation sample.

B Additional Tables

Table A1: Characteristics of Construction Sector Workers, 2004–2017

	Year			
	2004	2007	2012	2017
<i>Age distribution</i>				
Age <24	0.151	0.123	0.039	0.026
Age 24–35	0.450	0.445	0.360	0.231
Age 35–45	0.242	0.268	0.369	0.412
Age >45	0.157	0.165	0.232	0.330
Average age	34.02	34.73	38.24	40.93
<i>Education</i>				
Below secondary	0.758	0.746	0.656	0.671
Secondary	0.156	0.161	0.199	0.189
Tertiary	0.086	0.092	0.146	0.141
<i>Contract characteristics</i>				
Part-time	0.037	0.035	0.074	0.088
Fixed-term	0.724	0.659	0.472	0.508
<i>Demographics</i>				
Foreign born	0.163	0.275	0.182	0.184
<i>Occupational skill group</i>				
Very high skilled	0.019	0.021	0.046	0.040
High skilled	0.044	0.047	0.079	0.070
Medium-high skilled	0.054	0.056	0.086	0.074
Medium-low skilled	0.590	0.608	0.636	0.647
Low skilled	0.292	0.267	0.153	0.168

Notes: The table presents key characteristics of workers in the construction sector for selected years. Sample restricted to workers aged 18–60 employed in construction in the reference year.

Table A2: Heterogeneity in Cumulative Earnings Losses: Shock Interactions

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Baseline effect</i>						
Shock	−1.577*** (0.369)	−1.798*** (0.264)	−1.261*** (0.287)	−1.264*** (0.346)	−2.449*** (0.384)	−0.961*** (0.329)
<i>Panel B: Interaction with worker characteristics</i>						
Shock × Low education	−0.617** (0.295)					
Shock × Low tenure		−0.703* (0.381)				
Shock × Low experience			−1.838*** (0.282)			
Shock × Low earnings				−1.668*** (0.450)		
Shock × Low skill					0.491 (0.352)	
Shock × Young						−2.159*** (0.341)
Shock × Middle age						−0.514* (0.260)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Controls	✓	✓	✓	✓	✓	✓

Notes: Each column reports OLS estimates augmenting equation (2) with an interaction between the shock and an indicator for the low group within each dimension. Panel A reports the shock coefficient for the reference (high) group. Panel B reports the differential effect for the low group; the total effect for the low group is the sum of the two panels. Reference groups are: secondary education and above (Column 1), tenure above one year (Column 2), experience above six years (Column 3), above-median pre-recession earnings (Column 4), high-skilled (Column 5), and above 40 years of age (Column 6). Controls follow the same exclusion logic as Figure A16: the dimension being tested is excluded from the control set in the corresponding column. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Workers' wage and employment trajectories, 2003-2006

	Cumulative earnings			Avg. Earn-ings	Employment
	(1)	(2)	(3)	(4)	(5)
Pre-recession construction growth	15.71 (33.16)	5.239 (30.65)	-131.3 (135.5)	-4.426 (4.521)	0.122 (0.330)
Observations	30,253	30,253	30,253	30,253	30,253
Age $\times$ Gender $\times$ Education	✓	✓	✓	✓	✓
Job characteristics		✓	✓	✓	✓
Regional controls			✓	✓	✓

Notes: The dependent variables are computed over the pre-recession period 2003–2006: cumulative earnings, average annual earnings, and cumulative days worked. The regressor of interest is the instrument—average annual construction sector growth over 2000–2006—which in the main specification instruments for the provincial employment shock. Control variables follow the Table 1. Standard errors clustered at the province level. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table A4: First stage, pre-recession construction growth on construction sector decline by province, 2007-2012

	(1)	(2)	(3)	(4)
	Construction sector decline, 2007-2012			
Pre-recession construction growth	1.287*** (0.365)	1.274*** (0.369)	1.290*** (0.367)	1.209*** (0.374)
Observations	34,000	34,000	34,000	34,000

Notes: The main regressor is the instrument—average annual growth in construction employment over 2000–2006—on the provincial employment shock, defined as the proportional change in the construction sector's employment share between 2007 and 2012. Control variables follow Table 1. Column (1) reports the baseline specification. Column (2) adds average provincial growth in mortgage values over 2003–2006. Column (3) instead includes average growth in the value of new mortgages over 2003–2006. Column (4) adds average provincial growth in the immigrant population over 2000–2006. Standard errors are clustered at the province level. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table A5: Reallocation Index and Earnings Losses: OLS and IV Estimates

	Employment		Cumulative Earnings	
	(1) OLS	(2) IV	(3) OLS	(4) IV
Shock	−1.616*** (0.156)	−2.212*** (0.461)	−2.045*** (0.278)	−3.180*** (0.747)
Shock $\times$ Reall. Index	0.332*** (0.080)	0.570*** (0.147)	0.344** (0.162)	0.703*** (0.264)
Observations	34,000	34,000	34,000	34,000

Notes: Dependent variable is cumulative normalised earnings, 2007–2012. The reallocation index measures the expected earnings value of transitioning out of construction, standardised to mean zero and unit variance. The shock is the province-level decline in construction employment share, 2007–2012. Columns (2) and (4) instrument the shock with average annual construction employment growth, 2000–2006. Column (4) additionally instruments the shock $\times$ index interaction with pre-recession growth $\times$ index. All specifications include individual, contract, and regional controls. Clustered standard errors at the province level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: First-Stage Diagnostics

	Joint F $F(2, 49)$	SW F $F(1, 49)$
Shock	6.24 [0.004]	12.51
Shock $\times$ Reallocation index	19.63 [0.000]	39.96
Anderson-Rubin $F(2, 49)$:	11.84	[0.000]

Notes: First-stage results for the IV specification with two endogenous variables (Shock and Shock $\times$ Reallocation index) instrumented by pre-recession construction growth (2000–2006) and its interaction with the reallocation index. Joint F tests excluded instruments jointly for each endogenous variable. SW F is the Sanderson-Windmeijer conditional statistics, which test instrument relevance for each endogenous variable separately. All statistics are cluster-robust (50 provinces). p -values in brackets.

Table A7: Sectoral mobility and the reallocation index, 2012

	(1)	(2)	(3)
	Sector Change, 2012		
Shock	0.206** (0.081)	0.211*** (0.075)	
Reall. Index		-0.045* (0.024)	
Shock $\times$ Reall. Index		0.120*** (0.037)	
<i>Shock $\times$ Reallocation Index Quartile:</i>			
Q1 (lowest)			0.073 (0.110)
Q2			0.075 (0.130)
Q3			0.243*** (0.090)
Q4 (highest)			0.452*** (0.078)
Observations	32,521	32,521	32,521
Controls	Yes	Yes	Yes

Notes: Linear probability estimates. Dependent variable: indicator for employment in a sector other than construction 2012. All specifications include the preferred set of individual and regional controls from Column (3) of Table 1. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Wild Bootstrap Inference: Reallocation Index Specifications

	(1)	(2)	(3)
	Employment	Cumulative Earnings	Avg. Earnings
<i>Panel A: Additive specification</i>			
Shock	0.000	0.000	0.063
Reallocation index	0.020	0.000	0.007
<i>Panel B: Interaction specification</i>			
Shock	0.000	0.000	0.063
Reallocation index	0.002	0.361	0.855
Shock $\times$ index	0.000	0.071	0.665

Notes: Wild bootstrap p -values from 9,999 replications with Rademacher weights, clustering by province. Null imposed throughout. The wild bootstrap addresses finite-sample distortions with 50 province clusters.

Table A9: Cumulative earnings and employment Impact of the Construction Employment Decline, extended age and nativity

	Cumulative earnings			Employment		
	(1)	(2)	(3)	(4)	(5)	(6)
Shock	−2.062*** (0.293)	−1.942*** (0.286)	−2.016*** (0.291)	−1.625*** (0.160)	−1.598*** (0.162)	−1.508*** (0.170)
Observations	34,000	37,796	47,011	34,000	37,796	47,011

Notes: Cumulative earnings and cumulative days employed effect of the collapse in construction-sector employment. This specifications extend the baseline analysis by including older and foreign-born workers. Columns (1) and (4) replicate the baseline specification. Columns (2) and (5) expand the sample to include workers up to age 55. Columns (3) and (6) further include foreign-born workers. All specifications include the preferred set of individual and regional controls from Column (3) of Table 1. Robust standard errors are clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Robustness to pre-recession attachment thresholds

	(1) ≥ 3 months	(2) ≥ 9 months	(3) ≥ 12 months	(4) ≥ 3 months	(5) ≥ 9 months	(6) ≥ 12 months
	<i>Panel A. Earnings and employment outcomes</i>					
	Cumulative earnings			Employment		
	(1)	(2)	(3)	(4)	(5)	(6)
Shock	−2.067*** (0.259)	−2.026*** (0.307)	−2.032*** (0.312)	−1.614*** (0.138)	−1.694*** (0.168)	−1.699*** (0.183)
Observations	35,310	30,039	27,663	35,310	30,039	27,663
	<i>Panel B. Mobility outcomes</i>					
	Changed province			Changed sector		
Shock	−0.033 (0.116)	−0.011 (0.122)	0.008 (0.124)	0.352*** (0.050)	0.407*** (0.056)	0.453*** (0.056)
Observations	35,310	30,039	27,663	35,310	30,039	27,663

Notes: The table reports estimates of the effect of local shock exposure on cumulative outcomes over the post-2007 period under three alternative pre-recession attachment thresholds. Columns (1)–(3) and (4)–(6) successively require workers to have held employment for at least 3, 9, and 12 months in 2007. Panel A reports cumulative earnings and employment outcomes; Panel B reports the probability of changing province of residence and the probability of changing sector of employment. All specifications include the same control variables as in the main specification of Table 1. Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: First Post-Displacement Sector by Reallocation Index Quartile

Destination sector	Reallocation index quartile				Total
	Q1	Q2	Q3	Q4	
Agriculture	40.7	31.7	17.7	9.8	100.0
Extractive ind.	20.4	19.7	25.2	34.7	100.0
Manufacturing	22.7	22.4	28.1	26.8	100.0
Energy & utilities	21.9	20.3	27.4	30.4	100.0
Commerce	28.0	24.9	27.2	20.0	100.0
Hospitality	33.6	28.6	24.6	13.2	100.0
Transport & comm.	21.3	23.2	30.7	24.8	100.0
Finance & insurance	13.1	19.3	25.3	42.4	100.0
Real estate	26.1	24.3	30.4	19.1	100.0
Professional svc.	19.9	19.1	26.4	34.6	100.0
Public admin./educ.	22.4	24.5	25.9	27.2	100.0
Other services	20.0	10.9	38.2	30.9	100.0
Total	25.4	24.7	25.4	24.5	100.0

Notes: Each row reports the distribution across reallocation index quartiles of workers whose first post-displacement job was in the indicated sector. Entries are row percentages: each row sums to 100. The reallocation index quartiles are defined over the full estimation sample.

Table A12: Reallocation Index vs. Worker and Regional Characteristics

	Dependent variable: Employment					
	(1) Baseline	(2) + Educ	(3) + Skill	(4) + Exp	(5) + HHI	(6) + Unempl
Shock	-1.616*** (0.156)	-1.667*** (0.170)	-0.433 (0.585)	-2.211*** (0.213)	-1.143*** (0.419)	-1.234** (0.515)
Shock $\times$ RI	0.332*** (0.080)	0.317*** (0.103)	0.290* (0.158)	0.267*** (0.081)	0.308*** (0.076)	0.308*** (0.075)
Observations	34,000	34,000	34,000	34,000	34,000	34,000

Notes: All specifications replicate the baseline employment specification in Table 3. Columns (2)–(6) further allow the effect of the shock to vary with workers' education, skill group, potential experience, local labor market concentration (HHI), and the provincial unemployment rate in 2006 by interacting each characteristic with the shock measure. Standard errors are clustered at the province level and reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Robustness to Province Fixed Effects

	Employment		Earnings	
	(1) Baseline	(2) Province FE	(3) Baseline	(4) Province FE
Shock	−1.616*** (0.156)		−2.045*** (0.278)	
Shock $\times RI^w$	0.332*** (0.080)	0.322*** (0.070)	0.344** (0.162)	0.219 (0.146)
Observations	34,000	34,000	34,000	34,000
Province FE	No	Yes	No	Yes
Baseline controls	Yes	Yes	Yes	Yes

Notes: All specifications replicate the baseline employment specification in Table 3. Dependent variables are cumulative years of employment (columns 1–2) and normalized cumulative earnings (columns 3–4), both measured over 2007–2012. Province fixed effects in columns (2) and (4) absorb the shock main effect, which is therefore not identified and omitted. Standard errors are clustered at the province level and reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.1 Reallocation Index: Theoretical Framework

This appendix develops the search model that motivates the reallocation index of Section 4. Job seekers allocate search effort across local sectors, firms post vacancies subject to congestion, and local sectoral composition is fixed by predetermined production technology. The model rationalizes the index’s functional form, delivers the predictions tested in the body, and establishes that the index equals the model-implied rate at which a worker draws acceptable offers.

B.1.1 Environment

Time is continuous. The economy has R provinces indexed by r , each with J sectors indexed by j , including construction (cs). At $t = 0$, a unit mass of workers in province r is displaced from construction by a permanent contraction in construction demand, which I assume is unanticipated. Workers do not relocate across provinces, consistent with Section 6.2.²⁶

Worker i has observable skills $X_i \in \mathcal{X}$ determining productivity across sectors:

$$y_j(X_i) = a_j \cdot \phi_j(X_i), \quad (8)$$

where $a_j > 0$ is sector-specific TFP and $\phi_j(X_i) \in [0, 1]$ is match quality, interpreted as the

²⁶The no-migration assumption matches the empirical pattern; it is not a structural prediction of the model.

Table A14: Competing Risks: Immediate vs. Non-Immediate Reemployment

	Immediate reemployment			Non-immediate reemployment		
	(1)	(2)	(3)	(4)	(5)	(6)
RI^w	1.026*** (0.009)	0.818*** (0.041)		0.919*** (0.022)	0.670*** (0.101)	
Shock		1.160 (0.113)			1.708* (0.492)	
Shock $\times$ RI^w		1.487*** (0.127)			1.748** (0.422)	
Shock $\times$ Q1			0.891 (0.160)			1.232 (0.576)
Shock $\times$ Q2			1.018 (0.192)			0.547 (0.291)
Shock $\times$ Q3			1.223 (0.180)			1.903 (0.800)
Shock $\times$ Q4			1.906*** (0.409)			3.286** (1.904)
Observations	34,000	34,000	34,000	34,000	34,000	34,000
Failures	18,433	18,433	18,433	3,252	3,252	3,252
Displacement cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Shock $\times$ RI^w	No	Yes	Quartile	No	Yes	Quartile

Notes: Cause-specific Cox proportional hazard ratios reported throughout. $HR > 1$ indicates faster reemployment relative to the baseline group. Immediate reemployment is defined as reemployment in the same calendar month as displacement (spell length = 1 month); non-immediate reemployment requires at least one full calendar month of non-employment. Workers not reemployed by December 2012 are right-censored. Displacement cohort fixed effects (year of last construction employment, 2007–2012) are included in all specifications to absorb mechanical differences in the observation window: high-index workers exit construction later on average, shortening their window before the censoring date. Columns (1) and (4) include RI^w as a main effect only. Columns (2) and (5) add the shock main effect and the Shock $\times$ RI^w interaction. Columns (3) and (6) replace the continuous interaction with shock interactions for each quartile of the RI^w distribution. Efron method for tied failures. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

probability that the worker's skills meet the sector's task requirements.

Sector j in province r posts vacancies v_{jr} under free entry. Matching follows a Cobb-Douglas technology with sector-specific congestion,

$$m_{jr} = m_0 \cdot u_{jr}^{1-\eta} \cdot v_{jr}^\eta, \quad (9)$$

where u_{jr} is the measure of workers searching in sector j of province r , $\eta \in (0, 1)$ the vacancy elasticity, and $m_0 > 0$ baseline matching efficiency. The offer-receipt rate for a worker searching in sector j is

$$p_{jr} = \frac{m_{jr}}{u_{jr}} = m_0 \cdot \theta_{jr}^\eta, \quad (10)$$

with tightness $\theta_{jr} = v_{jr}/u_{jr}$.

Let $\sigma_j(X_i, r) \in [0, 1]$ denote the share of search effort worker i directs to sector j . The flow rate of acceptable offers from sector j is

$$\lambda_{ij}(r) = \sigma_j(X_i, r) \cdot p_{jr} \cdot \mathbb{P}(\text{accept} \mid \text{offer}, j, X_i), \quad (11)$$

and the worker accepts if $W_j(X_i) \geq U(X_i, r)$, where W_j and U are the values of employment in sector j and of unemployment. Unemployed workers receive flow utility b ; employed workers earn a Nash-bargained wage with worker power $\beta \in (0, 1)$:

$$w_{jr}(X_i) = \beta \cdot y_j(X_i) + (1 - \beta) \cdot b. \quad (12)$$

With job destruction rate $\delta > 0$ and discount rate $\rho > 0$, the employment value is

$$\rho W_j(X_i) = w_{jr}(X_i) + \delta [U(X_i, r) - W_j(X_i)]. \quad (13)$$

Firms pay flow cost κ_j per vacancy; free entry requires $\kappa_j = q_{jr} \cdot J_j(X_i)$, where $q_{jr} = m_0 \cdot \theta_{jr}^{\eta-1}$ is the vacancy-filling rate and $J_j(X_i)$ the firm's value of a filled position.

B.1.2 Equilibrium Search Allocation

The worker chooses $\{\sigma_j(X_i, r)\}_{j \neq cs}$ to maximize the value of unemployment:²⁷

$$\rho U(X_i, r) = b + \max_{\{\sigma_j\}} \sum_{j \neq cs} \sigma_j(X_i, r) \cdot p_{jr} \cdot \max\{W_j(X_i) - U(X_i, r), 0\}. \quad (14)$$

²⁷Search is restricted to non-construction sectors, reflecting the permanent contraction in local construction demand.

The objective is linear in $\{\sigma_j\}$, so without search costs the optimum concentrates all effort on the highest-return sector. A quadratic search cost $\chi(\sigma_j) = \sigma_j^2/2$ delivers interior allocations and a closed form.²⁸ Since sectors enter separably, the first-order condition gives

$$\sigma_j^*(X_i, r) = p_{jr} \cdot S_j(X_i), \quad (15)$$

where $S_j(X_i) = W_j(X_i) - U(X_i, r)$.

B.1.3 The Reallocation Index

Three functional-form restrictions, adopted for tractability, reduce the model to the index of Section 4.²⁹

Restriction A1 (linear surplus in match quality).

$$S_j(X_i) \approx \frac{\beta}{\rho + \delta} \cdot a_j \cdot \phi_j(X_i), \quad (16)$$

a first-order approximation around average match quality, obtained by substituting (12) and (8) into the value function.

Restriction A2 (tightness reflects local specialization).

$$\theta_{jr}^* = \left(\frac{m_0 \cdot \bar{S}_j}{\kappa_j} \right)^{1/(1-\eta)} \cdot \frac{s_{jr}}{\bar{s}_j}, \quad (17)$$

where $\bar{S}_j$ is average sector- j surplus and $s_{jr}/\bar{s}_j$ is local specialization. Combining free entry with matching (9), sectors with greater local specialization attract proportionally more vacancy posting.

Restriction A3 (acceptance scales with relative wages). The share of received offers accepted is proportional to w_j/w_{cs} , from aggregating acceptance decisions over reservation wages spanning the sectoral wage distribution.

²⁸Each σ_j is chosen independently, so the first-order condition yields $\sigma_j^* = p_{jr} \cdot S_j(X_i)$ without a Lagrange multiplier; imposing $\sum_j \sigma_j = 1$ would couple the conditions and preclude a closed form. The σ_j^* need not sum to one; the index is standardized in the empirical implementation.

²⁹Derivations are in Section A.2 of the Online Appendix.

Reallocation index. The flow rate of acceptable offers from sector j is $\lambda_{ij}(r) = \sigma_j^*(X_i, r) \cdot p_{jr} \cdot \alpha_j(X_i)$. Summing across sectors and substituting A1–A3 yields³⁰

$$\Lambda_i(r) = \Lambda_0 \cdot \sum_{j \neq \text{cs}} \phi_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{\text{cs}}}, \quad (18)$$

where Λ_0 absorbs m_0 , β , ρ , δ , κ_j , and the search-cost parameter and is common across workers and provinces. Normalizing by Λ_0 recovers the index of equation (1):

$$\text{RI}^w(X_i, r) \equiv \frac{\Lambda_i(r)}{\Lambda_0} = \sum_{j \neq \text{cs}} \phi_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{\text{cs}}}. \quad (19)$$

The three components map directly to the frictions identified in Section 4: skill compatibility, local vacancy intensity, and acceptance probability. They enter multiplicatively because, under A1–A3, these forces are non-separable determinants of the offer arrival rate.

Destination distribution. The probability that worker i reemploys in sector j is proportional to the acceptable-offer rate from that sector:

$$\pi_j(X_i, r) = \frac{\phi_j(X_i) \cdot (s_{jr}/\bar{s}_j) \cdot (w_j/w_{\text{cs}})}{\sum_{k \neq \text{cs}} \phi_k(X_i) \cdot (s_{kr}/\bar{s}_k) \cdot (w_k/w_{\text{cs}})}. \quad (20)$$

Two workers with identical RI^w share the same reemployment hazard but may differ in destination composition, one concentrating reemployment in a few well-matched sectors and another spreading it across more locally available options.

B.1.4 Testable Predictions

Proposition 1 (Predictions). *Under the environment of Sections B.1.1–B.1.3 and Restrictions A1–A3: (i) expected non-employment duration is strictly decreasing in $\text{RI}^w(X_i, r)$; (ii) the sectoral transition hazard is proportional to $\text{RI}^w(X_i, r)$; (iii) expected cumulative earnings losses over any post-displacement horizon are decreasing in $\text{RI}^w(X_i, r)$.*

Proof. (i) Conditional on $\{\sigma_j^*\}$ and $\{\theta_{jr}^*\}$, acceptable offers arrive as a Poisson process with rate $\Lambda_i(r) = \Lambda_0 \cdot \text{RI}^w(X_i, r)$, so expected non-employment duration is $1/[\Lambda_0 \cdot \text{RI}^w(X_i, r)]$, decreasing in the index. (ii) The sectoral transition hazard equals $\Lambda_i(r)$ by construction, depending on (X_i, r) only through the index. (iii) Cumulative losses combine foregone earnings during non-employment—governed by the index as in (i)—and the reemployment wage

³⁰Full derivations are in Online Appendix A.2. The reduction is not exact: province-invariant, sector-specific constants $\tilde{B}_j \cdot A_j$ are absorbed into the alternative-specific intercepts of the multinomial logit used to estimate $\phi_j(X_i)$.

gap. The expected reemployment wage $\sum_j \pi_j(X_i, r) w_j$ depends on (X_i, r) through the index components via (20). The wage component is approximate: the derivation assumes reservation wages are distributed around w_{cs} , and the approximation is tighter the smaller the dispersion of reservation wages across workers. $\square$

B.2 Extensions of the Baseline Framework

The baseline index corresponds to the equilibrium offer arrival rate under two restrictions calibrated to the Spanish construction collapse: limited geographic mobility ($\omega = 1$) and the effective disappearance of construction as a source of new employment ($\gamma = 0$), both supported empirically in Sections 2 and 6.2. Each can be relaxed while preserving the index's modular structure.

When workers can search in both the home province r and a neighboring province $\tilde{r}$, the index extends to:

$$\text{RI}^w(X_i, r, \tilde{r}) = \sum_{j \neq cs} \phi_j(X_i) \cdot \frac{\omega \cdot s_{jr} + (1 - \omega) \cdot s_{j\tilde{r}}}{\bar{s}_j} \cdot \frac{w_j}{w_{cs}}, \quad (21)$$

where $\omega \in [0, 1]$ weights the home province and $\omega = 1$ recovers the baseline. Geographic mobility enters exclusively through the local opportunity component $s_{jr}/\bar{s}_j$; the skill–demand match $\phi_j(X_i)$ and wage-acceptance term w_j/w_{cs} are unaffected. Geographic access and skill–demand alignment are therefore substitutes: workers with low home-province alignment can partially compensate by drawing on a better-matched neighboring market. The parameter ω is identified by cross-regional variation in bilateral mobility costs and could be estimated in settings where geographic adjustment is quantitatively important, trade-exposed commuting zones being a natural application.

When the contraction is incomplete, construction retains a fraction $\gamma \in [0, 1)$ of its pre-shock vacancy rate, and the index becomes:

$$\text{RI}^w(X_i, r, \gamma) = \sum_j \phi_j(X_i) \cdot \frac{s_{jr}(\gamma)}{\bar{s}_j} \cdot \frac{w_j}{w_{cs}}, \quad (22)$$

where $s_{cs,r}(\gamma) = \gamma \cdot s_{cs,r}$ and the summation includes construction; $\gamma = 0$ recovers the baseline. As γ rises, workers with high pre-shock construction attachment experience a smaller decline in effective search opportunities, and the index converges toward a conventional outside-options measure. The $\gamma \approx 0$ restriction is an implication of the free-entry condition: a permanent contraction implies a proportional decline in vacancy posting as expected profits fall below κ_j .

Unlike the outside-options index of [Caldwell and Danieli \(2024\)](#), which is a portable worker attribute, the reallocation index is a worker-in-province object: two workers with identical X_i face different prospects when local sectoral compositions differ (Section 6). The partial-shutdown extension is most relevant for gradual sectoral transitions—energy, automation—where γ can be proxied by the sector’s vacancy rate trajectory and identified from cross-worker variation in $\phi_{cs}(X_i)$ conditional on the aggregate shock.

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A Online Appendix

A.1 Reallocation Index: Construction Details

Sector-specific employment probabilities are estimated on a sample of non-construction workers drawn from the MCVL between 2002 and 2006. The estimation sample excludes public-sector employees and is restricted to workers observed at the start of a new employment spell, so that predicted probabilities reflect the characteristics associated with entering a job rather than remaining in one. For each of the twelve non-construction sectors, I estimate a multinomial logit model in which the outcome is the sector of employment. The covariate vector X_i includes interactions of four-year age categories with educational attainment, age categories with gender, a foreign-born indicator, and occupational skill group. The estimated coefficients are then applied to the characteristics of construction workers to obtain predicted employment probabilities $\hat{\phi}_j(X_i)$ for each sector j .

Let s_{jr} denote sector j 's employment share in province r in 2006, computed excluding construction workers, and let $\bar{s}_j$ denote the corresponding national share. The reallocation index for worker i in province r is:

$$RI(X_i, r) = \sum_{j \neq cs} \hat{\phi}_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{cs}}, \quad (\text{A1})$$

where w_j/w_{cs} is the ratio of sector j 's national median wage to the construction median, both measured in 2006. Local employment shares are normalized by their national counterparts so that the index captures relative provincial specialization rather than absolute sector size. The wage ratio downweights sectors offering substantially lower pay than construction, proxying for the acceptance margin in a standard job search model. The index is normalized to mean zero and unit standard deviation in the estimation sample to facilitate interpretation of regression coefficients.

Figure A3 shows the distribution of the normalized index among construction workers in 2007. The within-province standard deviation is approximately 2.3 times the between-province standard deviation, indicating that the index varies substantially across workers within the same local labor market. Identification of the reallocation effect therefore relies primarily on comparing workers in the same province rather than on cross-province differences in labor demand.

A.2 Additional Theoretical results

Derivation of Restriction A1. The Bellman equation for a worker employed in sector j at wage $w_{jr}(X_i)$ is

$$\rho W_j(X_i) = w_{jr}(X_i) + \delta [U(X_i, r) - W_j(X_i)], \quad (\text{A2})$$

where jobs are destroyed at rate δ and the worker returns to unemployment with value $U(X_i, r)$. Solving for $W_j(X_i)$:

$$W_j(X_i) = \frac{w_{jr}(X_i) + \delta U(X_i, r)}{\rho + \delta}. \quad (\text{A3})$$

Defining $S_j(X_i) \equiv W_j(X_i) - U(X_i, r)$ and substituting in (A3):

$$S_j(X_i) = \frac{w_{jr}(X_i) - \rho U(X_i, r)}{\rho + \delta}. \quad (\text{A4})$$

The Nash bargaining wage (12) and productivity function (8) give $w_{jr}(X_i) = \beta a_j \phi_j(X_i) + (1 - \beta) b$, so

$$S_j(X_i) = \frac{\beta a_j \phi_j(X_i) + (1 - \beta) b - \rho U(X_i, r)}{\rho + \delta}. \quad (\text{A5})$$

Equation (A5) is not yet linear in $\phi_j(X_i)$ because $U(X_i, r)$ depends on the full distribution of surpluses through (14):

$$\rho U(X_i, r) = b + \max_{\{\sigma_j\}} \sum_{j \neq cs} \sigma_j(X_i, r) \cdot p_{jr} \cdot S_j(X_i). \quad (\text{A6})$$

The surplus S_j thus appears on both sides of the system, precluding a closed-form solution without further restriction.

Let $\bar{\phi}_j$ denote average match quality and $\bar{U}(r)$ the unemployment value evaluated at the average. A first-order Taylor expansion of (A5) around $\phi_j(X_i) = \bar{\phi}_j$ gives

$$S_j(X_i) \approx \underbrace{\frac{\beta a_j \bar{\phi}_j + (1 - \beta) b - \rho \bar{U}(r)}{\rho + \delta}}_{\bar{S}_j(r)} + \frac{\beta a_j}{\rho + \delta} [\phi_j(X_i) - \bar{\phi}_j]. \quad (\text{A7})$$

The second-order correction is proportional to $\beta a_j / (\rho + \delta)$ times an equilibrium search-allocation weight and is therefore negligible for small deviations $\phi_j(X_i) - \bar{\phi}_j$. The intercept $\bar{S}_j(r) - \beta a_j \bar{\phi}_j / (\rho + \delta)$ varies across sectors but is constant across workers within a sector, depending only on average match quality $\bar{\phi}_j$ and the province-level unemployment value $\bar{U}(r)$; it shifts all workers' valuations of sector j equally and drops out when ranking sectors

by surplus. The surplus is therefore proportional to match quality:

$$S_j(X_i) \approx \frac{\beta}{\rho + \delta} \cdot a_j \cdot \phi_j(X_i). \quad (\text{A8})$$

Derivation of Restriction A2. The free-entry condition equates the flow cost of a vacancy to the expected return from filling it. From Nash bargaining the firm captures fraction $(1 - \beta)$ of the average match surplus $\bar{S}_j$, so the firm value of a filled position is $J_j = (1 - \beta) \bar{S}_j$. Substituting the vacancy-filling rate $q_{jr} = m_0 \theta_{jr}^{\eta-1}$ into the free entry condition and solving for tightness:

$$\theta_{jr}^{\text{nat}} = \left(\frac{m_0 (1 - \beta) \bar{S}_j}{\kappa_j} \right)^{1/(1-\eta)}. \quad (\text{A9})$$

This expression is province-invariant, capturing only the national-level attractiveness of sector j and implying uniform vacancy posting across provinces. Vacancy posting is instead more attractive where sector j is locally concentrated, since a thicker pool of suitable workers raises the expected return to posting. Let s_{jr} denote the employment share of sector j in province r and $\bar{s}_j$ its national counterpart, so that $s_{jr}/\bar{s}_j > 1$ identifies provinces specialized in sector j .³¹ Incorporating local specialization into the free-entry condition yields equilibrium tightness:

$$\theta_{jr}^* = \left(\frac{m_0 (1 - \beta) \bar{S}_j}{\kappa_j} \right)^{1/(1-\eta)} \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right), \quad (\text{A10})$$

Substituting (A10) into the offer-receipt probability $p_{jr} = m_0 \theta_{jr}^\eta$:

$$p_{jr} = m_0 \underbrace{\left(\frac{m_0 (1 - \beta) \bar{S}_j}{\kappa_j} \right)^{\eta/(1-\eta)}}_{A_j} \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta, \quad (\text{A11})$$

where A_j is a sector-specific constant absorbing all province-invariant parameters. The offer probability is proportional to $(s_{jr}/\bar{s}_j)^\eta$, so sectors with greater local specialization generate faster offer arrivals.

Derivation of Restriction A3. Worker i accepts a sector- j offer if $W_j(X_i) \geq U(X_i, r)$, equivalently if $S_j(X_i) \geq 0$. With reservation wages $w_i^R = \mu + \varepsilon_i$ and $\varepsilon_i \sim F$, the aggregate acceptance probability for a sector- j offer is

$$\alpha_j(X_i) = F(w_j - \mu). \quad (\text{A12})$$

³¹The assumption requires only that $s_{jr}/\bar{s}_j$ enters the free-entry condition monotonically, any functional form preserving this property yields the same equilibrium tightness expression.

Under a uniform distribution of ε_i on $[\underline{\varepsilon}, \bar{\varepsilon}]$ with support spanning the sectoral wage distribution, $\alpha_j(X_i)$ is linear in w_j .³² Normalizing by the pre-displacement wage w_{cs} gives

$$\alpha_j(X_i) \propto \frac{w_j}{w_{cs}}, \quad (\text{A13})$$

so sectors paying closer to pre-displacement earnings attract higher acceptance rates, capturing the wage-adequacy margin of the reemployment problem.

A.2.1 Derivation of the Reallocation Index

The flow rate of acceptable offers from sector j for worker i is the product of three terms: search intensity directed at sector j , the offer arrival rate conditional on search, and the probability that a received offer is acceptable:

$$\lambda_{ij}(r) = \sigma_j^*(X_i, r) \cdot p_{jr} \cdot \alpha_j(X_i). \quad (\text{A14})$$

Summing across sectors gives the total arrival rate of acceptable offers:

$$\Lambda_i(r) = \sum_{j \neq cs} \sigma_j^*(X_i, r) \cdot p_{jr} \cdot \alpha_j(X_i). \quad (\text{A15})$$

Under Restriction A3, the acceptance probability is proportional to the wage ratio: $\alpha_j(X_i) \propto w_j/w_{cs}$. Substituting into (A15):

$$\Lambda_i(r) = \sum_{j \neq cs} \sigma_j^*(X_i, r) \cdot p_{jr} \cdot \frac{w_j}{w_{cs}}. \quad (\text{A16})$$

Under Restriction A2, equilibrium tightness satisfies $\theta_{jr}^* = C_j \cdot (s_{jr}/\bar{s}_j)$, where $C_j = (m_0 \bar{S}_j / \kappa_j)^{1/(1-\eta)}$ collects national-level sector constants. The offer arrival rate is $p_{jr} = m_0 \cdot \theta_{jr}^\eta$, so:

$$p_{jr} = m_0 \cdot C_j^\eta \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta \equiv A_j \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta, \quad (\text{A17})$$

where $A_j \equiv m_0 \cdot C_j^\eta$ is province-invariant. Substituting into (A16):

$$\Lambda_i(r) = \sum_{j \neq cs} \sigma_j^*(X_i, r) \cdot A_j \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta \cdot \frac{w_j}{w_{cs}}. \quad (\text{A18})$$

Under Restriction A1, $S_j(X_i) \approx \frac{\beta}{\rho+\delta} a_j \phi_j(X_i)$. The optimal search intensity is

³²Monotonicity of α_j in w_j holds for any continuous F . The uniform specializes this to linearity, yielding the proportionality in equation (A13) without additional free parameters.

$\sigma_j^*(X_i, r) = p_{jr} \cdot S_j(X_i)$, so substituting A1 and A2:

$$\sigma_j^*(X_i, r) = A_j \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta \cdot \frac{\beta}{\rho + \delta} \cdot a_j \cdot \phi_j(X_i) \equiv \tilde{B}_j \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^\eta \cdot \phi_j(X_i), \quad (\text{A19})$$

where $\tilde{B}_j \equiv A_j \cdot a_j \cdot \beta / (\rho + \delta)$ is province-invariant. Substituting (A19) into (A18):

$$\Lambda_i(r) = \sum_{j \neq \text{cs}} \tilde{B}_j \cdot A_j \cdot \left(\frac{s_{jr}}{\bar{s}_j} \right)^{2\eta} \cdot \phi_j(X_i) \cdot \frac{w_j}{w_{\text{cs}}}. \quad (\text{A20})$$

The term $\tilde{B}_j \cdot A_j$ varies across sectors but is common across provinces. In the empirical implementation, sector-specific constants are absorbed into the alternative-specific intercepts of the multinomial logit used to estimate $\phi_j(X_i)$, so they cancel out of the cross-worker and cross-province comparisons that identify the index.

The specialization ratio enters with exponent 2η . Setting $\eta = 1/2$ delivers $2\eta = 1$, so the ratio enters linearly. Any $\eta \in (0, 1)$ preserves monotonicity of the index in local specialization, so this restriction affects the curvature of the relationship but not its direction. [Petrongolo and Pissarides \(2001\)](#) find that estimates of η cluster between 0.5 and 0.7 across a wide range of countries and time periods, making $\eta = 1/2$ an empirically grounded baseline. The sector-specific constants $\tilde{B}_j \cdot A_j$, which are province-invariant, are absorbed into the alternative-specific intercepts of the multinomial logit used to estimate $\phi_j(X_i)$ and play the role of Λ_0 in the empirical implementation. The arrival rate therefore reduces to:

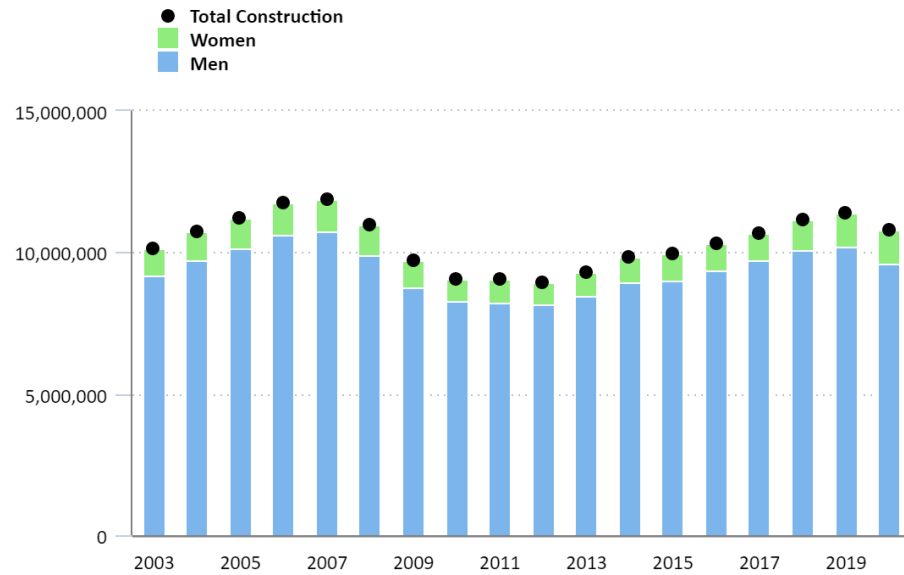
$$\Lambda_i(r) = \Lambda_0 \cdot \sum_{j \neq \text{cs}} \phi_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{\text{cs}}}, \quad (\text{A21})$$

and the reallocation index is defined as the model-implied arrival rate normalized by Λ_0 :

$$\text{RI}^w(X_i, r) \equiv \frac{\Lambda_i(r)}{\Lambda_0} = \sum_{j \neq \text{cs}} \phi_j(X_i) \cdot \frac{s_{jr}}{\bar{s}_j} \cdot \frac{w_j}{w_{\text{cs}}}. \quad (\text{A22})$$

A.3 Additional Figures

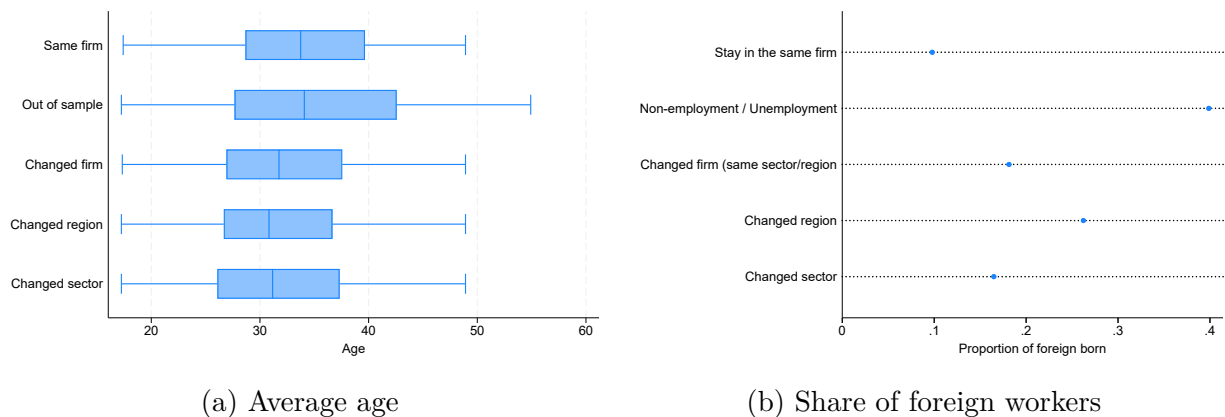
Figure A4: Construction sector employment in the US, 2003–2020



Source: U.S. Bureau of Labor Statistics.

Notes: Construction employment by gender in the US, 2003–2020. Source: US Bureau of Labor Statistics.

Figure A5: Characteristics of workers initially in the construction sector by employment status in 2013



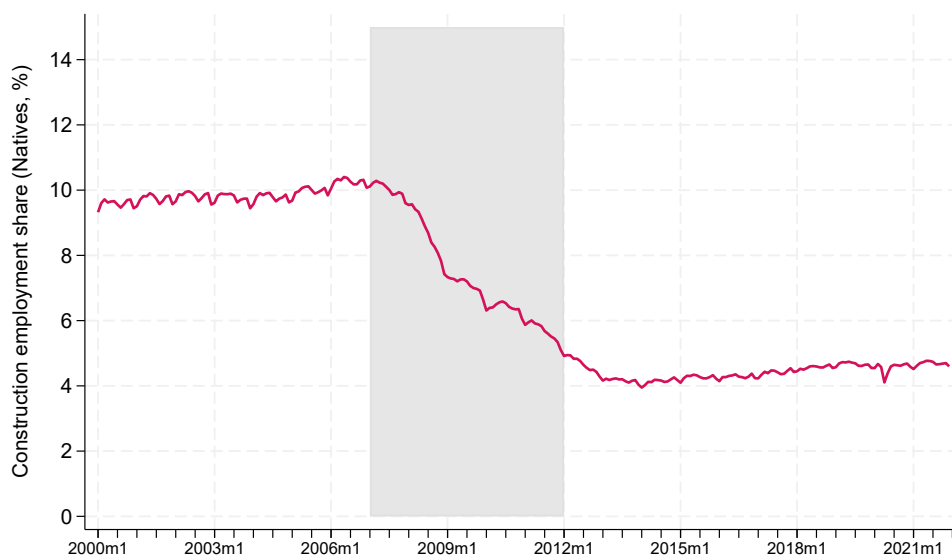
Notes: Panel (a) Average age in 2007 of workers in the construction sector by status in 2013. Panel (b) Share of foreign workers in the construction sector by status in 2012. The sample is restricted to workers in the construction sector in 2007 aged 20–55 years old.

Figure A6: Employment share of male workers in the construction sector, 1995–2019



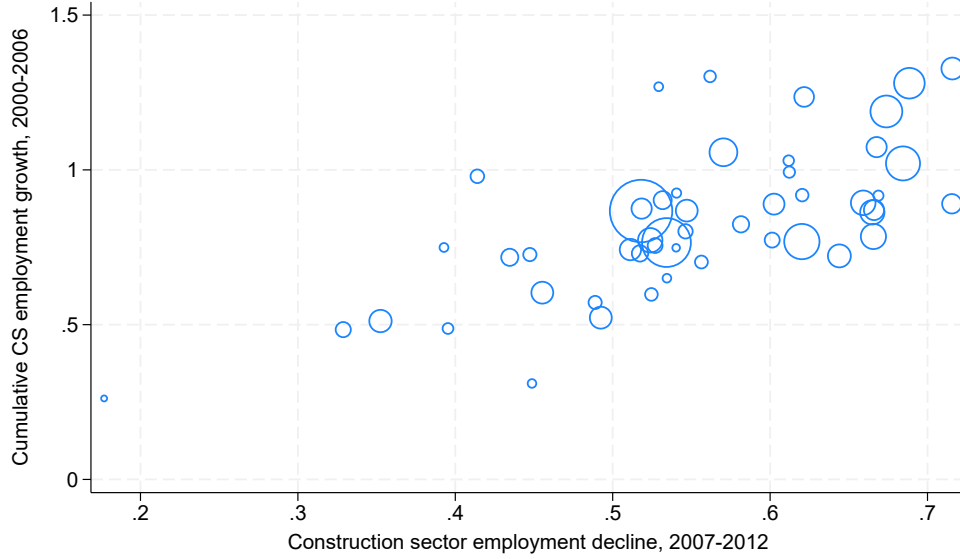
Notes: Presents the proportion of male workers in Spain's construction sector from January 2000 to December 2017. The data are restricted to male workers aged 20–60 and employed during the referenced period. The shaded area comprises the years of the Great Recession in Spain, between 2007 and 2012.

Figure A7: Employment share of native workers in the construction sector, 1995–2019



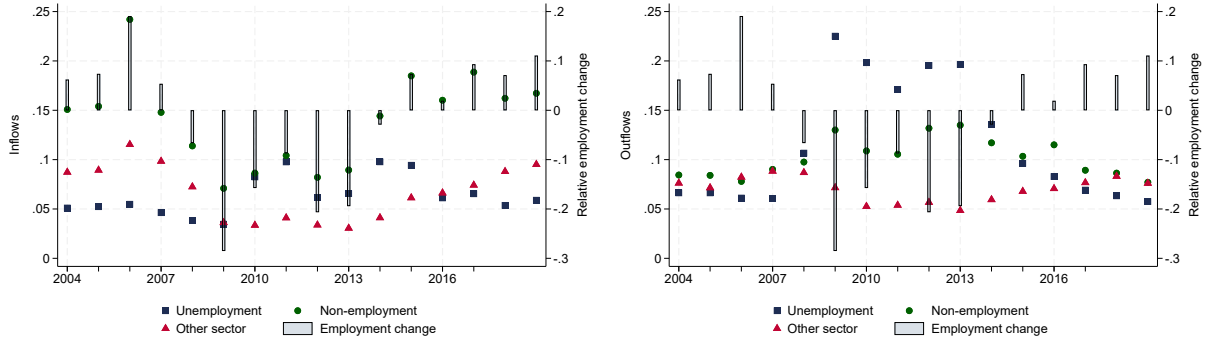
Notes: Presents the proportion of native workers in Spain's construction sector from January 2000 to December 2017. The data are restricted to native workers aged 20–60 and employed during the referenced period. The shaded area comprises the years of the Great Recession in Spain, between 2007 and 2012.

Figure A8: Cumulative growth in the construction sector, 2000-2006



Notes: Change in the employment share of the construction sector by province between 2007 and 2012 against cumulative employment share growth in the construction sector between 2000 and 2006 weighted by the employment in the construction sector in 2006.

Figure A9: Aggregate flows from/to the construction sector

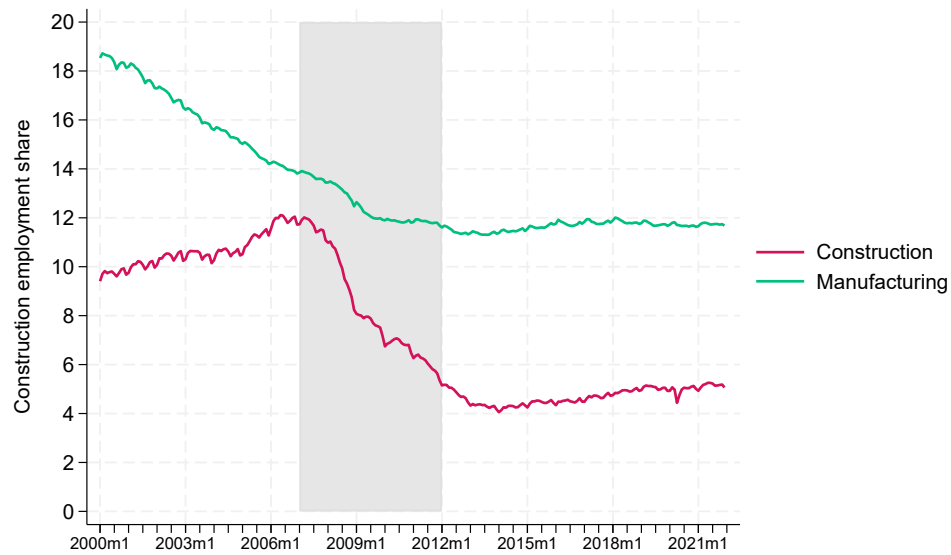


(a) Inflows

(b) Outflows

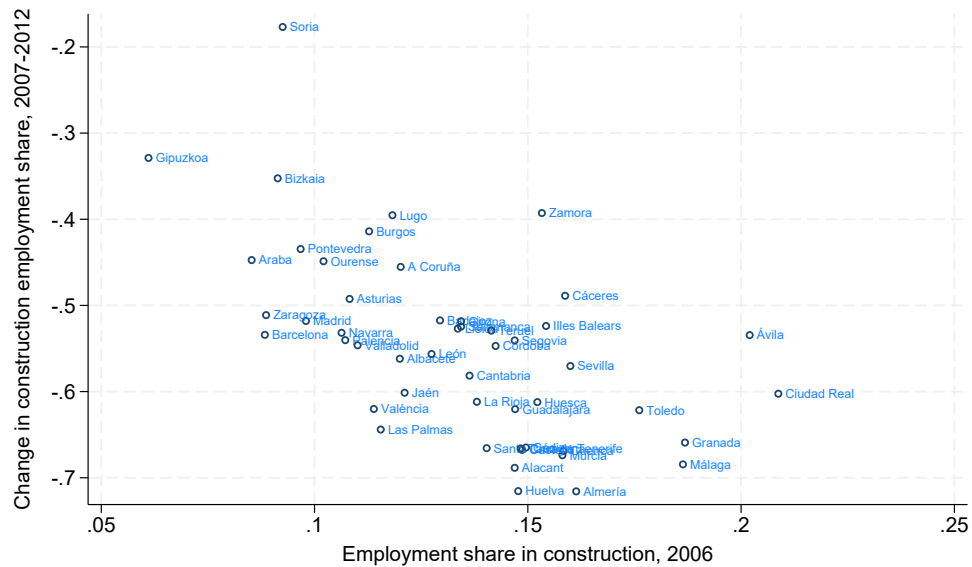
Notes: Panel (a) reports inflows into the construction sector, defined as the share of individuals who transition into construction from another sector, non-employment, or unemployment between years $t - 1$ and t , relative to total employment in year $t - 1$. Panel (b) reports outflows from the construction sector, defined as the share of workers employed in construction in year $t - 1$ who transition to another sector, non-employment, or unemployment in year t . The sample consists of annual observations from 2003 to 2019 for workers aged 18–60.

Figure A10: Manufacturing and construction employment shares, 1995–2019



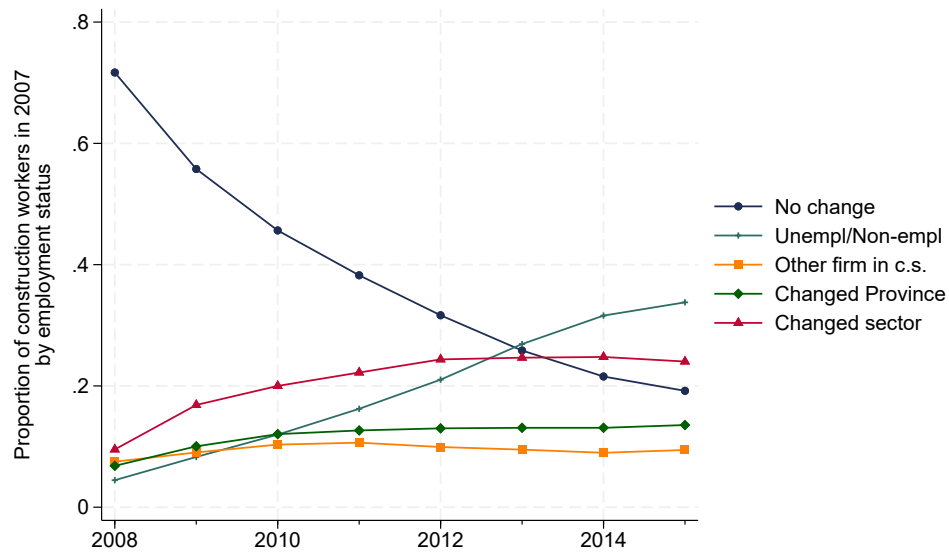
Notes: Employed workers in the construction and manufacturing sector as a percentage of all employed workers, January 1995 to December 2019. Data is restricted to workers employed during the referenced period. The shaded area comprises the years of the Great Recession in Spain, between 2008 and 2014.

Figure A11: Change in the construction employment share by province, 2003–2020



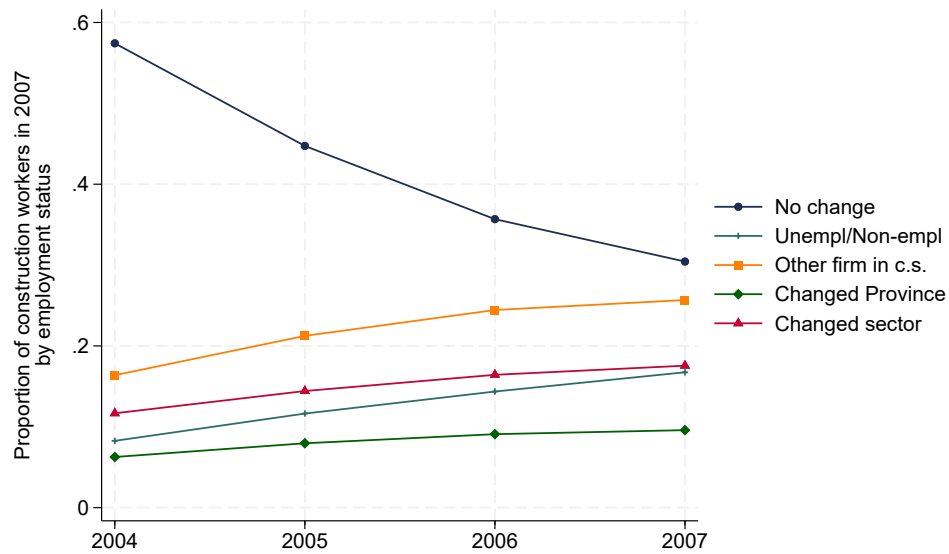
Notes: Proportional change in the employment share of the construction sector by province between 2007 and 2012 against construction employment share in 2006. The sample considers the 50 Spanish provinces.

Figure A12: Working status of highly skilled individuals in 2007



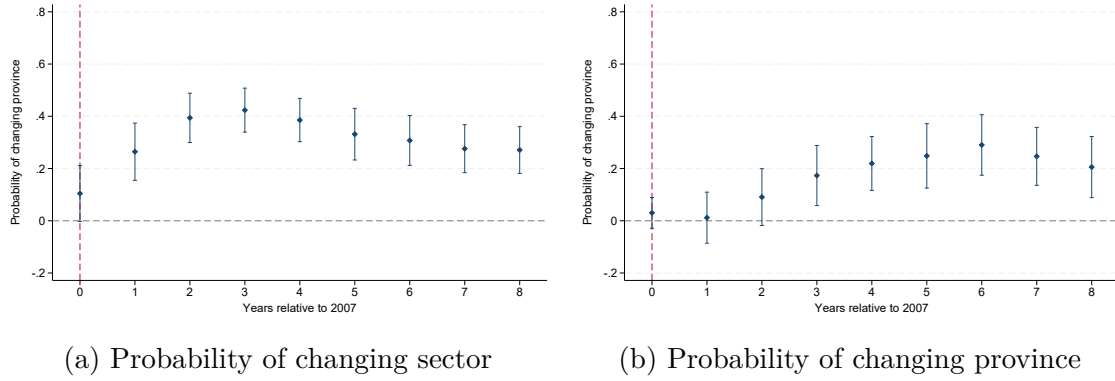
Notes: The shares are computed based on highly skilled workers, and every year, I tracked their working status up to 2015.

Figure A13: Working status of workers in the construction sector in 2003



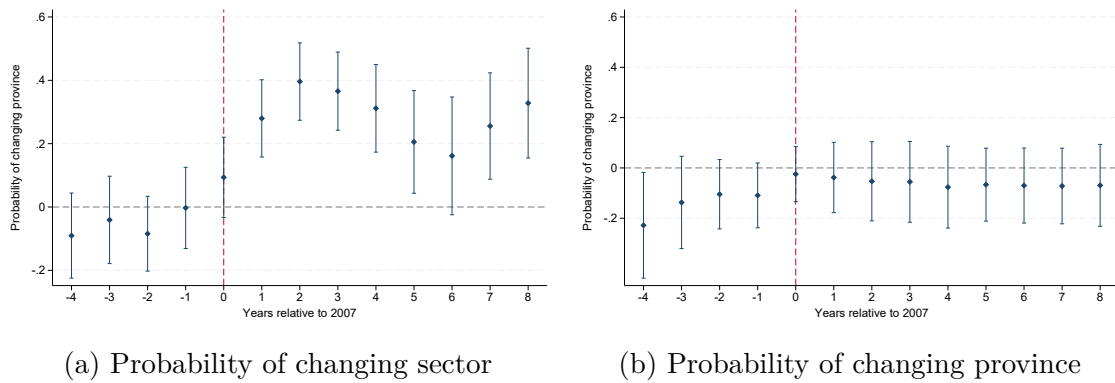
Notes: The shares are computed based on native workers employed in the construction sector in 2003, and every year I tracked their working status up to 2007.

Figure A14: Probability of Sectoral versus Geographic Reallocation



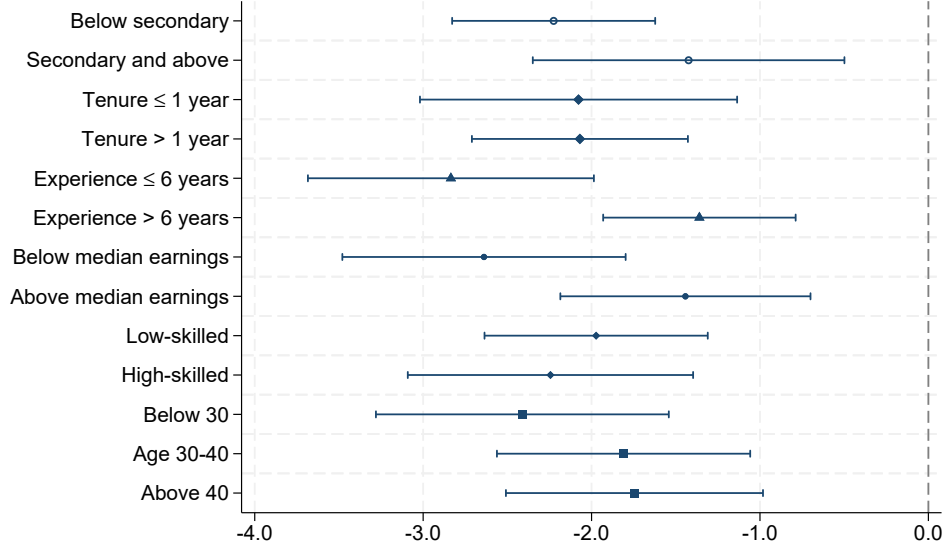
Notes: Sample restricted to workers aged 29–42 employed in construction in 2007. Panel (a): probability of transitioning to a different sector; Panel (b): probability of residing in a different province from the 2007 province of residence. Additional controls include the initial construction employment share, a Bartik-type demand shifter, and demographic characteristics. Cumulative change province or sector

Figure A15: Probability of Sectoral versus Geographic Reallocation



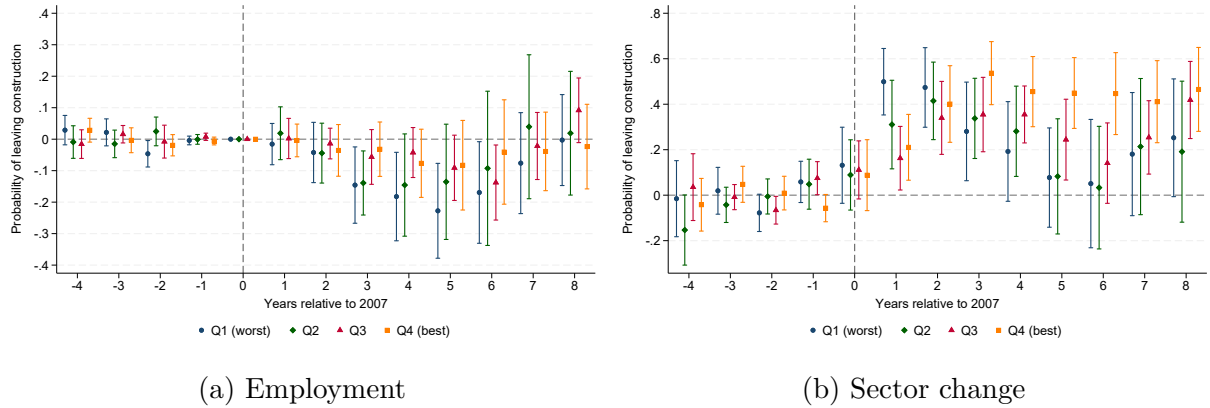
Notes: Panel (a): probability of transitioning to a different sector; Panel (b): probability of residing in a different province from the 2007 province of residence.

Figure A16: Heterogeneity of the Shock's Impact on Earnings Losses by Worker Characteristics



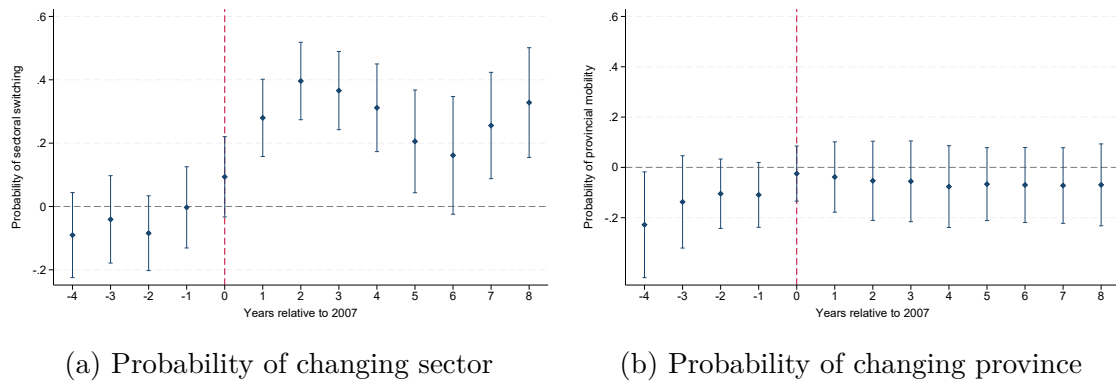
Notes: Estimated earnings losses by baseline worker and employment characteristics. Sample restrictions and control variables are identical to those used in Table 1. Standard errors are clustered at the province level.

Figure A17: Probability of Employment and Sectoral Exit by Reallocation Index Quartile



Notes: Year-by-year linear probability estimates. Dependent variable: indicator for employment in a sector other than construction in year t , for $t \in \{2003, \dots, 2015\}$. Each year is estimated separately as a regression of the annual sector-change indicator on the shock interacted with indicators for each quartile of the reallocation index, conditional on quartile fixed effects and the pre-shock (2003–2006) average of the sector-change indicator. Controls and sample restrictions follow Table 1. Standard errors clustered at the province level. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Figure A18: Probability of Sectoral versus Geographic Reallocation (Unconditional)



Notes: Panel (a) reports the probability of transitioning out of the construction sector into a different sector; Panel (b) reports the probability of residing in a province different from the 2007 province of residence. The sample and control variables follow the baseline specification in Table 1.

A.4 Additional Tables

Table A15: Descriptive Statistics, by Sector of Employment in 2007

	Non-construction	Construction
<i>Labor market outcomes</i>		
Cumulative earnings	61.56	45.80
Years employed	4.55	3.48
<i>Education</i>		
Below secondary	0.45	0.76
Secondary	0.26	0.16
Tertiary	0.29	0.08
<i>Worker characteristics</i>		
Tenure (years)	3.57	2.06
Age	33.60	32.54
Female	0.47	0.08
Foreign-born	0.14	0.28
Observations	304,085	52,671

Notes: Workers are classified by sector of employment in 2007. Cumulative earnings are defined as total non-zero earnings over 2007–2012 divided by average monthly earnings in 2005–2006. Education shares and gender and nativity shares sum to one within each column. Standard deviations in parentheses.